MILES & MEMORIES CO.

EVERY MILE BECOMES A MEMORY

CAPSTONE PROJECT
POWER BI | DAX | POWER QUERY

CASE STUDY

Miles & Memories Executive Dashboard

How I turned two and a half years of bike sales into six pages of clear decisions.



6
DASHBOARD PAGES

255
DAX MEASURES

27
TABLES IN THE MODEL

56,046
SALES ROWS ANALYSED

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Every Mile Becomes a Memory.

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WHERE TO FIND THE REQUIRED SUBMISSION ITEMS

Required item	Where it is in this case study	Page
1 Project overview	Sections 1 and 2	4
2 Dashboard screenshots (1920 x 1080, 16:9)	Section 14, all six pages	35
3 Data analysis and methodology	Sections 5 to 9 and the four core analyses	8
4 Key takeaways	Key takeaways, summary and conclusions	44
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7 Conclusions	Key takeaways, summary and conclusions	44
8 Recommendations	Section 15	43

The brand story

Where Miles & Memories Co. comes from

Miles & Memories Co. is a made up cycling brand. I created it for my Power BI capstone because I did not want to show a plain practice dataset. I wanted it to feel like a real company with real people waiting for answers.



My brand sheet. Logo, colour palette and visual direction, all made by me.

I used the AdventureWorks data from my data analytics bootcamp. I gave it a name, a logo, a colour palette and a story: cycling, adventure, and the moments that happen along the way.

Then I built the report around the parts of a business every leader worries about: revenue, products, profit, customers and growth. The goal was never to only draw charts. It was to turn data into clear actions.

MY COLOURS

Adventure Blue #1769C2
Deep Navy #0B2D4D
Sky Blue #6DB7E8
Ice White #F7FAFC
Slate #526575

MY FONT

Poppins for everything: headings, numbers and labels. One font keeps the whole brand clean and modern.

MY BRAND LINE

Every Mile Becomes a Memory.

1 Overview

What I built and why

I built a six page executive dashboard in Power BI for Miles & Memories Co. A leader can open it and answer one question on each page: how are we doing, what sells, who buys, where do we sell, where are we heading, and what should we do next.

$24.91M
total revenue, Jan 2021 to Jun 2023

$10.46M
gross profit at a 42% margin

25.16K
orders from 17.42K customers

$990
average order value

The six pages

Page	The question it answers
Overview	How big are we, and is growth turning into profit?
Products	What sells and what earns?
Customers	Who rides with us and who comes back?
Territory	Where are the miles being made?
Trends	Where is the road heading?
Strategy	What do we do next, and how much is it worth?

What is inside

- **7 source tables** loaded from CSV files with Power Query
- **6 relationships** in a star schema with one snowflake branch
- **255 DAX measures** and 15 tables made with DAX
- **A JSON theme** so every chart follows my brand
- **Bookmarks and buttons** for page navigation and toggles
- **What-if sliders** that estimate the value of each action

HOW TO READ THIS CASE STUDY

I wrote it in the order I worked: the problem, the data, the model, the numbers, the design, then what I found and what I would do about it. The full list of every DAX measure is at the end in Appendix A.

The dashboard covers the period from 1 January 2021 to 30 June 2023, which is 30 months of actual sales.

2 The business problem

Revenue is growing. But is the business getting healthier?

Miles & Memories Co. had a lot of data and very little clarity. Sales were going up, but nobody could say why, which customers mattered most, or where to put money and effort next.

What the leaders could not see

Growth without proof

Revenue jumped after July 2022, but profit margin slipped by 0.13 points. Growth did not mean better profit.

Money in too few places

Bikes are 94.89% of revenue. The top 10% of customers bring in 40.36%. One bad month in either place would hurt.

Many reports, no decisions

Numbers lived in separate files. There was no single page that said what to do on Monday morning.

My problem statement

> *"Where does our growth really come from, who and what is carrying us, and which actions will raise revenue and margin the most?"*

Why it matters

- Leaders need one trusted place to look, not ten spreadsheets.
- A growing top line can hide a weak margin, so profit must sit next to revenue on the first page.
- Every recommendation needs a number beside it, so the team can pick what to do first.

MY ROLE

I played the business intelligence analyst. I prepared the data, built the model, wrote the measures, designed the pages and wrote the recommendations. The numbers, captions and colours on the dashboard come from measures. The only text I wrote myself is the short “What this means” notes on each page.

3 Business questions

Twelve questions the dashboard has to answer

Before I drew a single chart, I wrote the questions. Each page exists to answer a few of them, and each answer ends in an action.

OVERVIEW

1. How much revenue and profit do we make, and is growth real?
2. Is growth turning into profit, or only into sales?
3. Which category, country and product carry us?

PRODUCTS

4. What sells the most, and what earns the most?
5. Which categories should we reprice, promote or expand?

CUSTOMERS

6. Who are our best customers and how much do they matter?
7. Are riders coming back, and who is slipping away?

TERRITORY

8. Which markets are biggest, and which are growing fastest?
9. What separates a strong market from a weak one?

TRENDS

10. Is growth sustained, and what does the next quarter look like?
11. Is there a best time of year to sell?

STRATEGY

12. What should we do next, and what is each action worth?

WHY I ASK QUESTIONS FIRST

A dashboard that starts with charts becomes decoration. A dashboard that starts with questions becomes a tool. When a chart did not help answer one of these twelve questions, I took it out.

4 The data set

AdventureWorks, loaded from CSV files

I used the AdventureWorks sample data from my bootcamp. It has seven tables: one table of sales and six tables that describe what was sold, who bought it, where, and when.

Table	Source	Rows	What is in it
Fact_Sales	3 yearly files, combined	56,046	One row per order line: dates, order number, product, customer, territory, quantity
Dim_Customers	1 CSV	18,148	Name, birth date, income, children, education, job, home owner
Dim_Products	1 CSV	293	Product name, model, colour, size, style, cost and price
Dim_Product_Subcategories	1 CSV	37	Subcategory name and its category key
Dim_Product_Categories	1 CSV	4	Bikes, Accessories, Clothing, Components
Dim_Territories	1 CSV	10	Region, country and continent
Dim_Calendar	1 CSV	911	One row per day, 1 Jan 2021 to 30 Jun 2023

What I noticed first

- **The sales table has no price.** It only has quantity. So revenue has to be quantity times the price from the product table.
- **Accessories and clothing only start in July 2022.** Before that, the company sold bikes only.
- **The data stops on 30 June 2023.** That date became my "today" for every calculation.

What it covers

- 6 countries: United States, Australia, United Kingdom, Germany, France and Canada
- 4 categories, 37 subcategories and 130 products that were actually sold
- 17,416 customers who placed at least one order

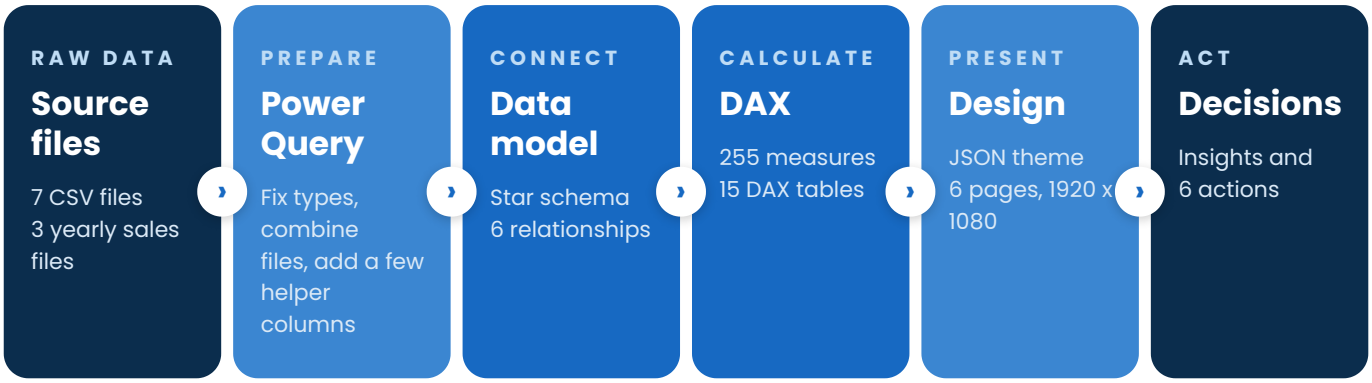
A NOTE ON THE STORY

The data is real sample data, but the company, brand and its goals are fictional. I made them up so the dashboard reads like a real business case.

5 How it was built

My pipeline from raw files to decisions

I worked in a straight line. Each step feeds the next one, and each step has one job.



What happens at each step

1. **Source files.** I pointed Power Query at the CSV files. The three yearly sales files sit in one folder, so new files can join the same flow.
2. **Power Query.** I changed data types, fixed text, combined the sales files into one table and added only the few columns that filter data cleanly.
3. **Data model.** I connected the tables with many-to-one relationships. Dimensions filter the fact table, never the other way around.
4. **DAX.** I wrote every number as a measure, so the dashboard recalculates for any filter. I also made small tables with DAX for customer segments, order baskets, buttons and sliders.
5. **Design.** I applied a JSON theme, built a sidebar menu, KPI cards and a "What this means" panel on every page, then named and grouped every visual.
6. **Decisions.** Each page ends with a recommendation and the Strategy page turns them into six actions with estimated value.

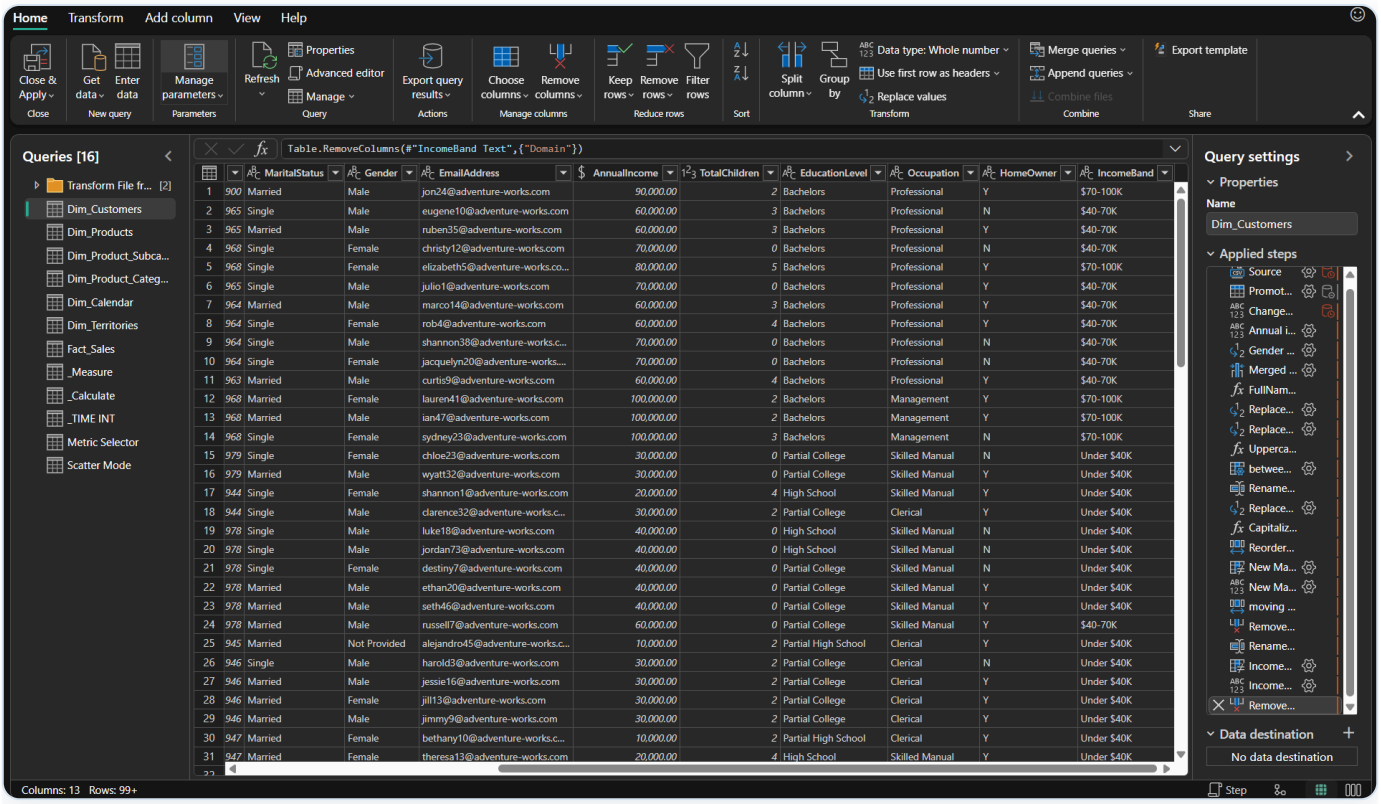
ONE RULE I KEPT

Raw data is cleaned once and kept light. Everything that answers a question is a DAX measure. That keeps the model fast and makes every number easy to trace back.

6 Data preparation and cleaning

Power Query: fix the formats, add the right columns

In Power Query I did the work that must happen before the model, such as fixing formats and combining files. I kept each query short and every step has a clear name, so anyone can read my steps and see what I did.



Dim_Customers in Power Query. Each applied step has its own name on the right, like "Gender Not Provided" and "Merged FullName".

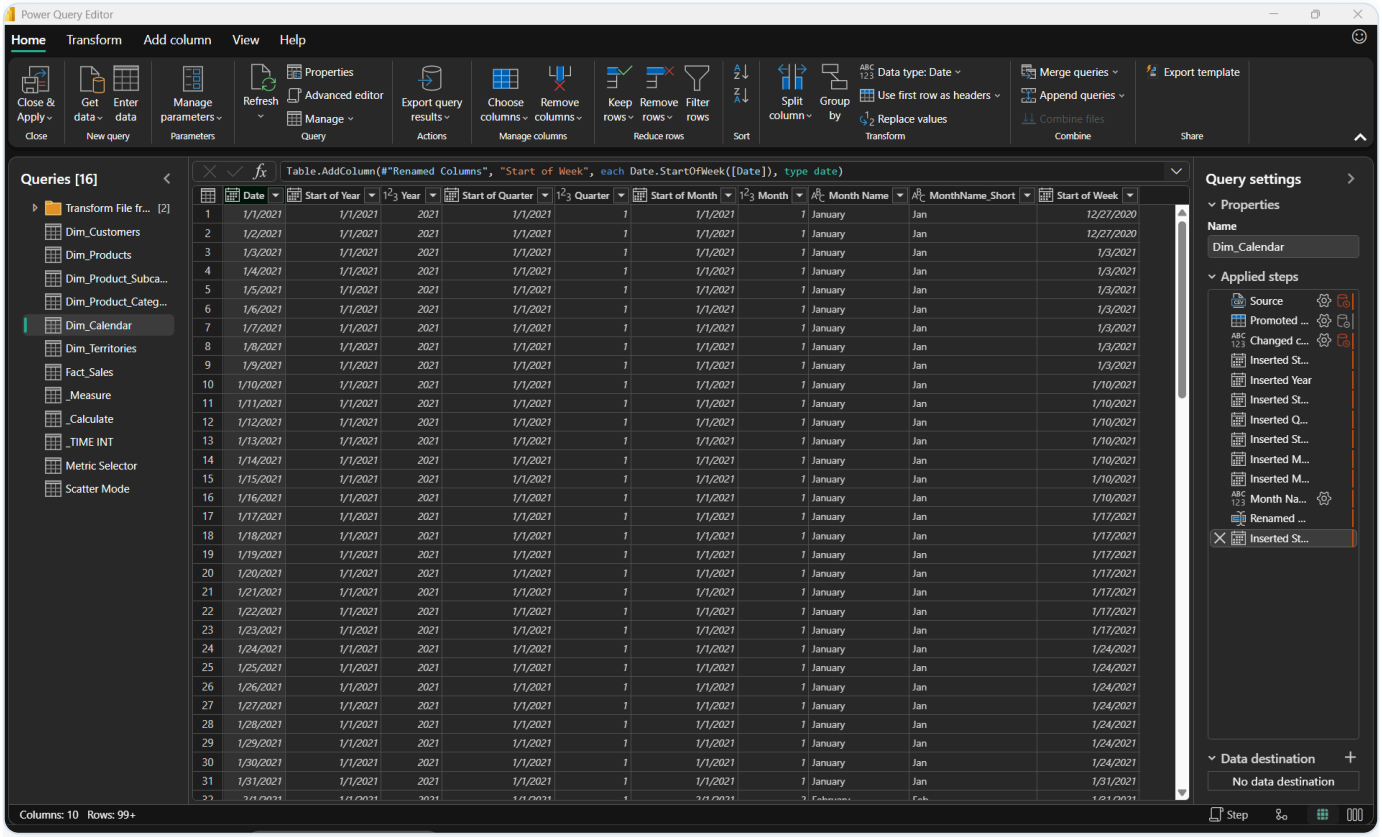
What I did to the customer table

- Set the right data types, such as a date for BirthDate and currency for AnnualIncome
- Replaced empty gender values with "Not Provided"
- Joined FirstName and LastName into one FullName
- Made names look neat with capital letters on each word
- Changed "M" and "F" to Male and Female
- Changed marital status letters to Married, Single or Other
- Added an **IncomeBand** column so customers can be grouped by income
- Removed helper columns I no longer needed

Power Query (M): the income band rule

```
IncomeBand = Table.AddColumn(#"Renamed New Marital Status", "IncomeBand",
    each if [AnnualIncome] <= 40000 then "Under $40K"
    else if [AnnualIncome] <= 70000 then "$40-70K"
    else if [AnnualIncome] <= 100000 then "$70-100K"
    else "$100K+")
```

Calendar and sales files



Dim_Calendar and the sales folder. The calendar steps add year, quarter, month, month name and week from a single Date column.

COMBINING THE YEARLY SALES FILES

I did not load each sales file by hand. I pointed Power Query at the **sales folder**. It reads every file the same way, joins them into one table, sets the data types, and drops the file name column. When a new year arrives, I drop in the file and refresh.

ADDING COLUMNS TO THE CALENDAR

From one Date column I made Start of Year, Year, Start of Quarter, Quarter, Start of Month, Month, Month Name, a short month name (Jan, Feb) and Start of Week.

Power Query (M): combine a folder of yearly files

```
Source = Folder.Files("...\Dataset\AW Sales"),
#"Read Each Sales File" = Table.AddColumn(..., "Transform File", each #"Transform File"([Content])),
#"Combined All Three Years" = Table.ExpandTableColumn(..., "Transform File", ...),
#"Set Correct Data Types" = Table.TransformColumnTypes(..., {{"OrderDate", type date}, ...})
```

WHY I CLEANED IN DAX INSTEAD OF POWER QUERY

I made a choice that I want to explain clearly: **less is more**. I could have added hundreds of calculated columns in Power Query, such as margin, growth and flags. But every column is stored in the model, takes memory and makes the file slower.

So I kept Power Query for the basics and wrote the rest as **DAX measures**. Measures take almost no space because they are calculated only when a visual asks for them. My model stays small and fast, and all my business logic lives in one place that is easy to find.

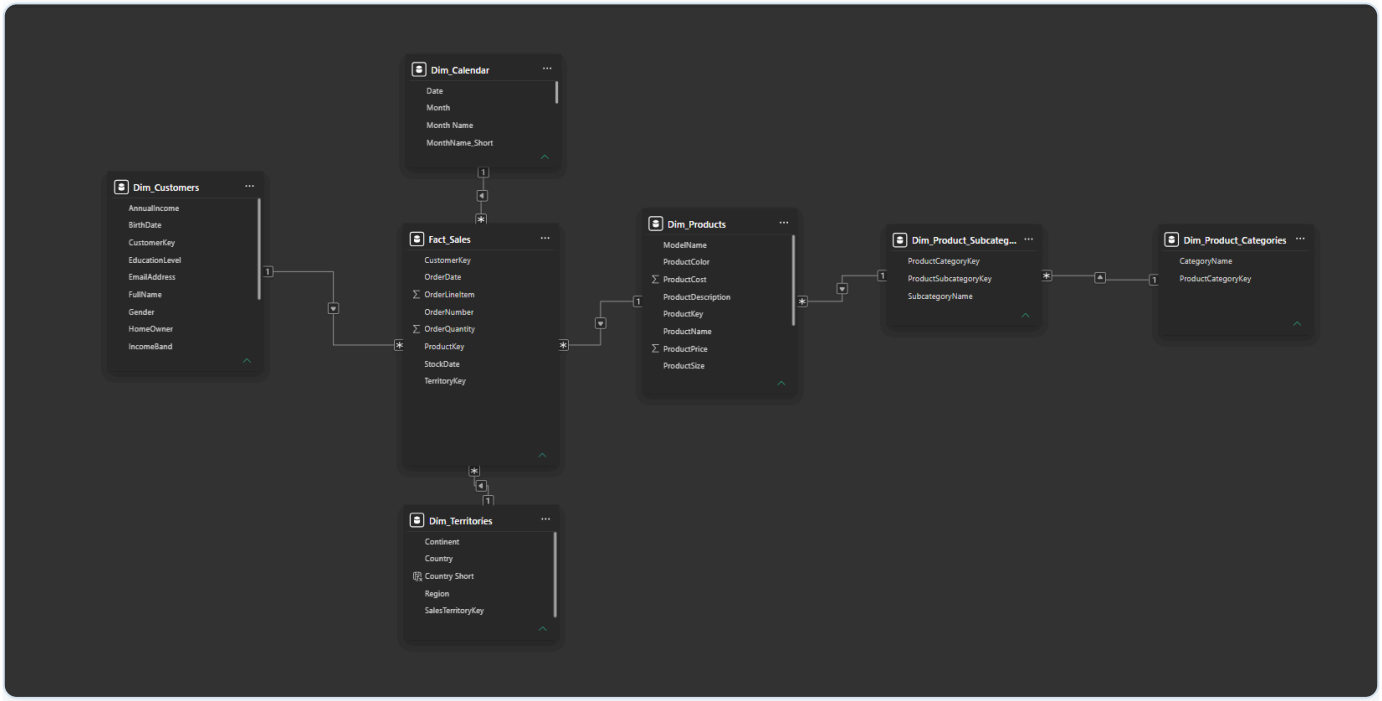
7
queries loaded

0
heavy calculated columns in the big sales table

255
measures instead, calculated only when needed

7 The data model

A star schema with one snowflake branch



My model view. Fact_Sales sits in the middle. Customers, Territories, Products and the Calendar connect to it. The product tables continue on to Subcategories and Categories.

A **star schema** puts one fact table (the events, here the sales) in the centre and the description tables (dimensions) around it like the points of a star. It is simple, fast and easy to understand.

Mine has one extra step: **Dim_Products → Dim_Product_Subcategories → Dim_Product_Categories**. A chain of dimension tables like this is called a **snowflake**. I kept it because it matches how the data came, and it lets me filter by category and still reach every product.

From (many)	To (one)	Type
`Fact_Sales[CustomerKey]`	`Dim_Customers[CustomerKey]`	Many to one, single direction
`Fact_Sales[TerritoryKey]`	`Dim_Territories[SalesTerritoryKey]`	Many to one, single direction
`Fact_Sales[ProductKey]`	`Dim_Products[ProductKey]`	Many to one, single direction
`Fact_Sales[OrderDate]`	`Dim_Calendar[Date]`	Many to one, single direction
`Dim_Products[ProductSubcategoryKey]`	`Dim_Product_Subcategories[...]`	Many to one, single direction
`Dim_Product_Subcategories[ProductCategoryKey]`	`Dim_Product_Categories[...]`	Many to one, single direction

WHY I BUILT IT THIS WAY

- **Many to one, single direction.** Filters flow from the description tables down to the sales. No loops and no surprises.
- **Keys only.** The sales table has just keys, dates and a quantity, so it stays narrow.
- **Price lives in Products.** Revenue uses RELATED to look up the price, so one price change updates everything.

TABLES WITH NO RELATIONSHIP, ON PURPOSE

- **Buttons and toggles** (Nav Page, View Toggle, Segment View, Metric Selector, Scatter Mode). They only hold labels and an order.
- **Sliders** (four what-if tables). They only hold numbers from 0 to 100.
- **Customer RFM and Order Basket.** Built from the sales, then used on their own for segments and baskets.

A CHOICE I EXPLAIN ON THE TRENDS PAGE

I did not mark Dim_Calendar as the date table. Because of that, my month over month measure uses REMOVEFILTERS and EDATE to find the month before, which works in any chart.

8 KPIs

The numbers every page starts with

A KPI is a number that tells a leader how the business is doing at a glance. I picked a small set, defined each one in plain words, and put them at the top of the pages where people look first.

KPI	Value	In plain words	Why I track it
Total Revenue	$24.91M	Quantity sold times product price, added up	How big the business is
Total Profit	$10.46M	Revenue minus cost	What is left after the product cost
Profit Margin %	41.97%	Profit divided by revenue	Whether growth is healthy
Total Orders	25.16K	Each order number counted once	How often people buy
Average Order Value	$990.09	Revenue divided by orders	How big each basket is
Total Customers	17.42K	Each customer counted once	How wide the base is
Repeat Customer Rate	33.63%	Customers with more than one order	Loyalty and coming back
Top Decile Share	40.36%	Revenue from the top 10% of customers	How concentrated the money is
MoM Growth	7.91%	This month against the month before	Momentum
Potential Revenue Opportunity	$2.01M	Sum of three gaps times the slider settings	What the next actions are worth

The base of almost everything

```
Total Revenue =
SUMX(
    Fact_Sales,
    Fact_Sales[OrderQuantity] *
RELATED(Dim_Products[ProductPrice])
)
```

DIVIDE avoids errors; + 0 shows 0, not blank

```
Profit Margin % =
DIVIDE( [Total Profit], [Total Revenue] ) + 0
```

I use DIVIDE and not the "/" sign so a missing value shows 0 and not an error.

HOW I USED THE KPIS

Every KPI card on the dashboard is a measure, not a typed number. If the date picker on top changes, all of them change together. The first card on each page is a dark blue card so the eye lands there first.

9 Calculated measures (DAX)

255 measures, organised so I can find any of them in seconds

All of my business logic is written as DAX measures. A measure is a formula that is worked out only when a chart asks for it, so it does not make the file bigger. That is why I put everything there and not into columns.

255
measures in the model

238
used by a visual or by another measure

17
leftovers I can delete

72
visuals read Active Page, my most used measure

Why 255 and not 25?

I did not set out to write 255 measures. The number grew because of one rule I kept: **numbers, captions and colours are never typed in**. A KPI is a measure. The words under it are a measure. The colour of a number is a measure. The label on a button is a measure. When the date picker moves, all of them move together. The only text I wrote myself is the short “What this means” notes on each page.

WHERE THE 255 COME FROM, BY WHAT THEY DO

Text and captions	Titles, subtitles and the "What this means" sentences that change with the data	92
Calculations	Gaps, baskets, segments, growth, rankings and the Strategy numbers	84
Time intelligence	Trailing 3 months, month over month, half year growth, forecast	37
Colour rules	Green or red, dark or light blue, chosen by a rule	15
Controls	Read which button or slider is selected	14
Base KPIs	Revenue, cost, profit, orders, customers and friends	13

How I organised them

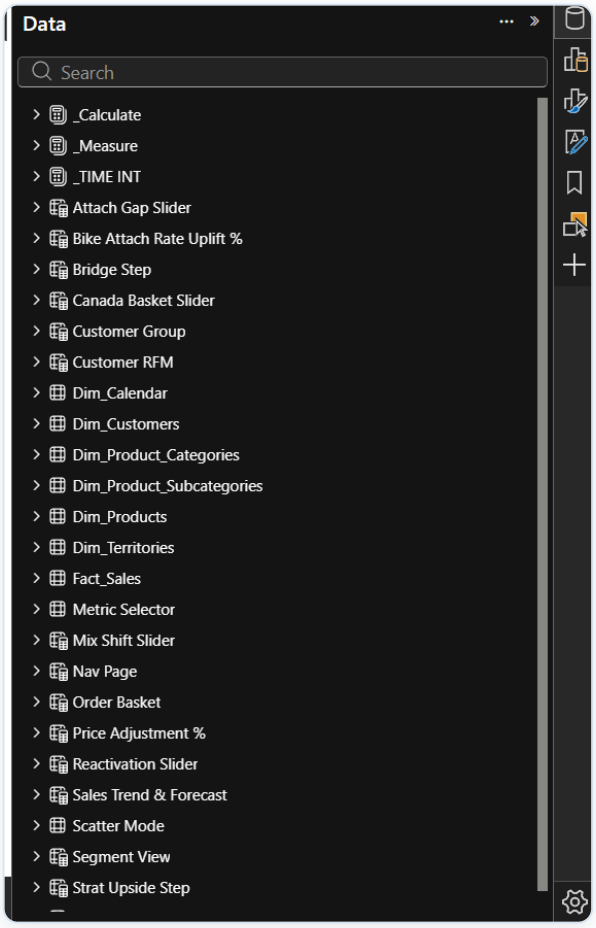
I put each measure in the table that matches its job, so the Data pane reads like a table of contents.

Table	Measures	What lives there
`_Measure`	124	Base KPIs, colour rules, text and captions, and button readers
`_Calculate`	84	Calculations with logic: gaps, baskets, segments, strategy
`_TIME INT`	37	Anything that compares periods: trailing, growth, forecast
`10 helper measures`	10	Sit in their own parameter or selector table, because they belong to that control

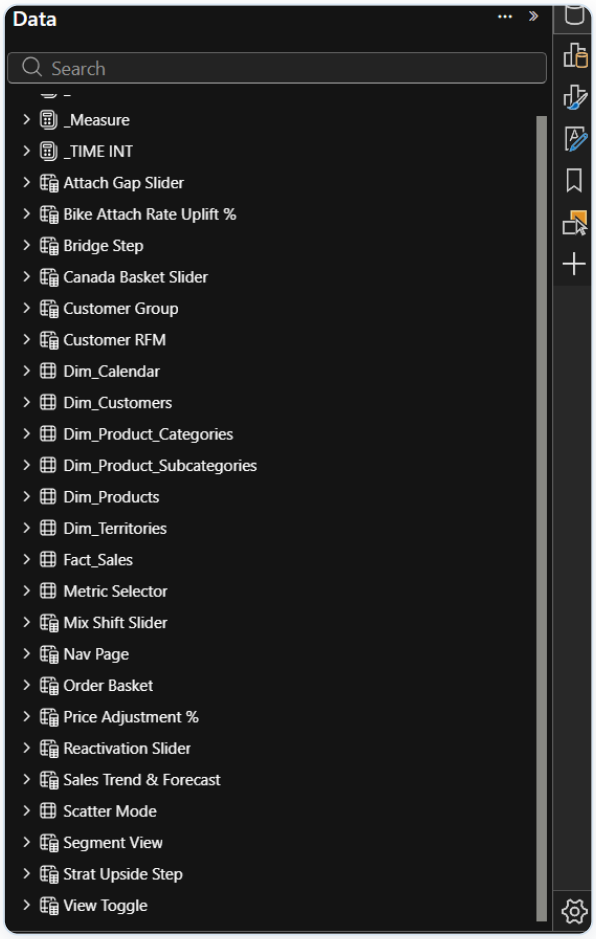
WHY I DID NOT LEAVE THEM ON THE SALES TABLE

I could have saved every measure on the table where it started. With 255 measures, that would bury my columns and make every formula hard to find. So I gave measures their own tables, and each table has one job.

- **Easy to find.** The underscore makes the three tables sort to the top of the Data pane. When I need a number, I know where to look before I scroll.
- **One job each. _Measure** answers "what is the number, the colour or the words?". **_Calculate** answers "what does the logic say?". **_TIME INT** answers "how does this compare with another period?". If a result looks wrong, I already know which table holds the formula.
- **Clean data tables.** The sales and dimension tables keep only their columns, so anyone adding a chart sees fields, not hundreds of formulas.
- **No duplicates.** Before I write a new measure, I check these three tables first and reuse what is there. That keeps the count low and the model fast.
- **Helpers stay with their control.** The 9 helper measures sit in the small table of the button or slider they serve, so if I delete that control the helper goes with it.



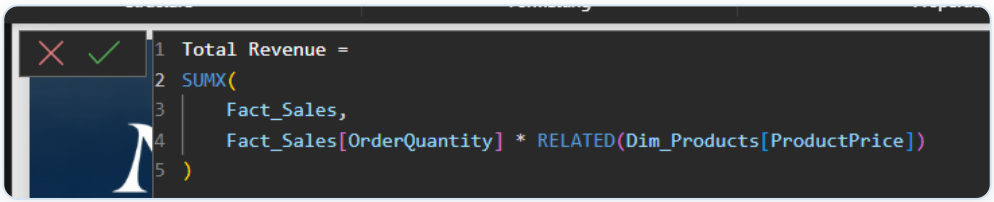
Data pane, top. The three measure tables come first.



Data pane, scrolled. Tables follow in A to Z order.

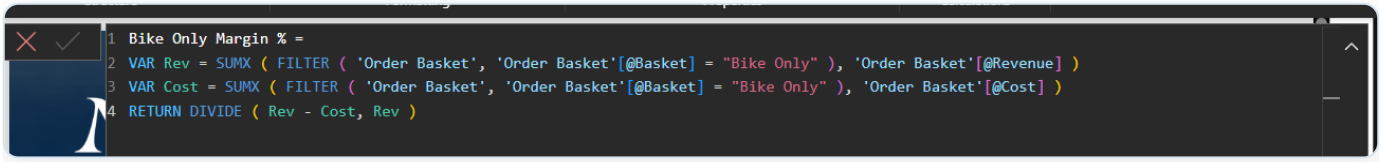
The building blocks

Almost every measure is built on a few simple ones. Once these four were right, the rest was easy. These are my own screenshots from the formula bar.



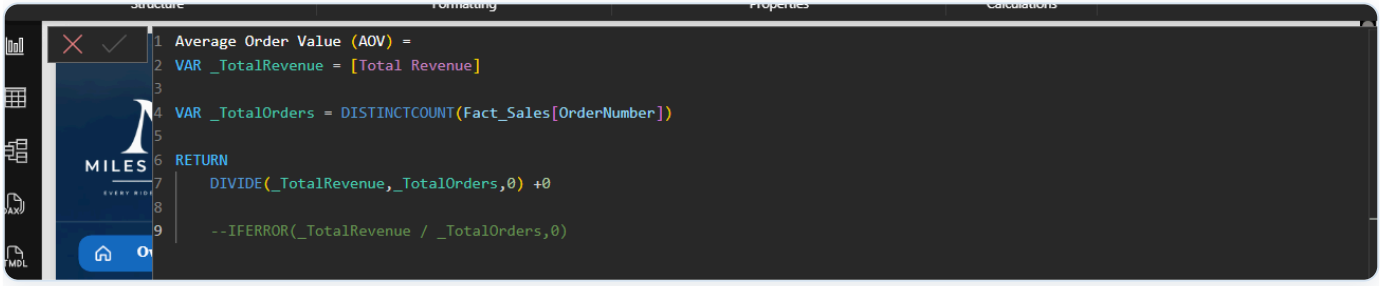
```
Total Revenue =
SUMX(
    Fact_Sales,
    Fact_Sales[OrderQuantity] * RELATED(Dim_Products[ProductPrice])
)
```

Total Revenue. There is no price in the sales table, so I multiply the quantity by the price found in Dim_Products with RELATED. This one measure is read by 68 other measures.



```
Bike Only Margin % =
VAR Rev = SUMX ( FILTER ( 'Order Basket', 'Order Basket'[@Basket] = "Bike Only" ), 'Order Basket'[@Revenue] )
VAR Cost = SUMX ( FILTER ( 'Order Basket', 'Order Basket'[@Basket] = "Bike Only" ), 'Order Basket'[@Cost] )
RETURN DIVIDE ( Rev - Cost, Rev )
```

Bike Only Margin %. First version, built on my Order Basket table (orders with a bike and no accessory). I later rewrote it to follow the dates (Appendix C).



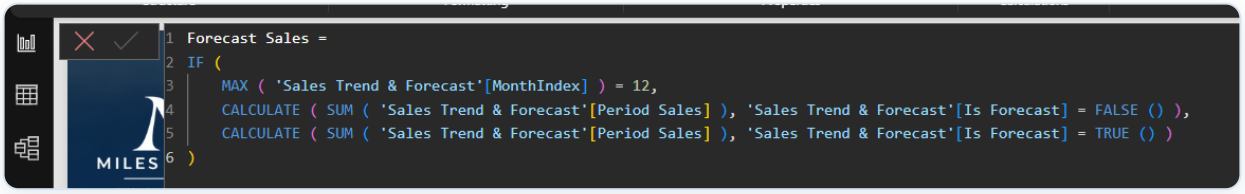
```
Average Order Value (AOV) =
VAR _TotalRevenue = [Total Revenue]

VAR _TotalOrders = DISTINCTCOUNT(Fact_Sales[OrderNumber])

RETURN
    DIVIDE(_TotalRevenue,_TotalOrders,0) +0

    --IFERROR(_TotalRevenue / _TotalOrders,0)
```

Average Order Value. I keep revenue and orders in two VAR boxes, then divide. The + 0 at the end shows 0 when there is no data.



```
Forecast Sales =
IF (
    MAX ( 'Sales Trend & Forecast'[MonthIndex] ) = 12,
    CALCULATE ( SUM ( 'Sales Trend & Forecast'[Period Sales] ), 'Sales Trend & Forecast'[Is Forecast] = FALSE () ),
    CALCULATE ( SUM ( 'Sales Trend & Forecast'[Period Sales] ), 'Sales Trend & Forecast'[Is Forecast] = TRUE () )
)
```

Forecast Sales. It picks the actual line for the first 12 months and the projection after that, so one measure can draw both parts of the line.

HOW I READ MY OWN FORMULAS

VAR is a named box that holds a number. RETURN is the answer. Writing a long formula as small boxes makes it easy to read and easy to fix when something looks wrong.

Trailing 3M Revenue (_TIME INT)

```
CALCULATE (
    [Total Revenue],
    DATESINPERIOD ( Dim_Calendar[Date], MAX (
Dim_Calendar[Date] ), -3, MONTH )
)
```

MoM Chart Growth % (_TIME INT)

```
VAR ThisMonth = MAX ( Dim_Calendar[Start of Month] )
VAR Cur = [Total Revenue]
VAR Prev =
    CALCULATE (
        [Total Revenue],
        REMOVEFILTERS ( Dim_Calendar ),
        Dim_Calendar[Start of Month] = EDATE ( ThisMonth,
-1 )
    )
RETURN
    IF ( ISBLANK ( Prev ), BLANK (), DIVIDE ( Cur - Prev,
Prev ) )
```

Trailing 3M looks back three months from the last date on the chart. Month over month uses REMOVEFILTERS and EDATE, so each bar finds the month before it by itself.

The measures behind my recommendations

The Strategy page runs on a handful of measures. Each one finds a **gap**, which is money the business is not collecting yet, and then the sliders decide how much of it to count.

Strat Basket Gap $ (_Calculate)

```
VAR Weak = [Strat Weak Market]
VAR WeakOrders = CALCULATE ( [Total Orders],
Dim_Territories[Country] = Weak )
VAR WeakAov = CALCULATE ( [Average Order Value (AOV)],
Dim_Territories[Country] = Weak )
RETURN
    WeakOrders * ( [Company AOV] - WeakAov )
```

Strat Attach Gap $ (_Calculate)

```
VAR Orders =
    ADDCOLUMNS (
        SUMMARIZE ( Fact_Sales,
            Fact_Sales[OrderNumber] ),
        "@HasBike", ..., "@HasAccessory", ... )
VAR BikeOnly = COUNTROWS ( FILTER (
    Orders, [@HasBike] && NOT [@HasAccessory] ) )
VAR AccBasket = ...  // same as before
RETURN
    BikeOnly * AccBasket
// Reads Fact_Sales, so it follows the dates.
```

Strat Revenue Upside (_Calculate)

```
[Strat Basket Gap $] * [Canada Basket Slider Value] /
100
    + [Strat Attach Gap $] * [Attach Gap Slider Value]
/ 100
    + [Strat Reactivation $] * [Reactivation Slider
Value] / 100
```

Strat Upside Value (_Calculate)

```
SWITCH (
    SELECTEDVALUE ( 'Strat Upside Step'[Order] ),
    1, [Total Revenue],
    2, [Strat Basket Gap $] * [Canada Basket Slider
Value] / 100,
    3, [Strat Attach Gap $] * [Attach Gap Slider
Value] / 100,
    4, [Strat Reactivation $] * [Reactivation Slider
Value] / 100
)
```

Rules that change the colour of a number

MoM Label Color (_Measure)

```
IF ( [MoM Chart Growth %] < 0, "#D94F5C", "#1E9E6B" )
```

Seasonality Bar Color (_Measure)

```
VAR Cur = [Total Revenue]
VAR Best = MAXX ( ALLSELECTED (
Dim_Calendar[MonthName_Short] ), [Total Revenue] )
RETURN
    IF ( Cur = Best, "#1769C2", "#6DB7E8" )
```

Customer and button measures

Top Decile Share % (Customer RFM)

```
DIVIDE ( CALCULATE ( SUM ( 'Customer RFM'[Monetary] ),
'Customer RFM'[Decile] = 1 ), SUM ( 'Customer
RFM'[Monetary] ) )
```

Active Page (_Measure)

```
SELECTEDVALUE ( 'Nav Page'[Order], 0 )
```

Category Action (_Calculate)

```
SWITCH (
        SELECTEDVALUE (
Dim_Product_Categories[CategoryName] ), "Bikes",
"Reprice", "Accessories", "Promote", "Clothing",
"Expand", "Components", "Dormant", "Maintain" )
```

Selected Metric (Metric Selector)

```
SELECTEDVALUE ( 'Metric Selector'[Metric], "Revenue" )
```

THE FULL LIST

Appendix A lists all 255 measures from most important to least important, grouped by the table they live in, with a one line plain English reason for each.

10 Tables, charts and how they connect

Every visual earns its place by answering a question

I picked every visual to answer one question. The **Table** visual works like an Excel pivot table: a field in rows, a measure in values and a filter on top. I used tables wherever a leader needs to read exact numbers, and charts wherever the shape tells the story.

Page	Visual	Why I chose it
Overview	Column and line chart	Is growth turning into profit? Bars show the month, the line smooths it so one spike does not fool us.
Overview	Donut and bar charts	Which category and which country carry us? Parts of a whole need a donut. Ranking needs bars.
Overview	Two stat tiles	Bike orders with and without an accessory, side by side, so the difference is obvious.
Products	Scatter (bubble) chart	Revenue against margin for each product. The four corners say Promote, Expand, Review and Reprice.
Products	Table with actions	Exact numbers for each category, with a Reprice, Promote, Expand or Dormant call.
Customers	Table (RFM segments)	Five customer groups with their size, spend, recency and the play to run.
Customers	Decile bars and stacked columns	How concentrated is revenue, and are new orders becoming returning orders?
Territory	Table, donut, bubble, column	Compare countries by size, basket and growth, because size alone is not the story.
Trends	Line with forecast	Actuals, a 3 month average and a dashed projection with a marked start.
Trends	Growth, seasonality, waterfall	Where momentum broke, which month is best, and what built the growth from H1 2022.
Strategy	Six action cards, sliders, waterfall	Turn each finding into an action with a value that moves when I drag a slider.

MY CHART RULES

- A **bar** for ranking, a **line** for time, a **donut** only for a few parts of a whole.
- **One message** per chart, written as a question in the title.
- **Blue first.** The most important bar or number is dark blue and the rest are sky blue.
- Red and green only when the meaning is good or bad.

WHY I ADDED A SENTENCE UNDER MANY CHARTS

A chart shows the shape. A sentence says what it means. Under each chart I added one line, written by a measure, so it always matches the numbers above it. This helps people who read quickly or who do not like to read charts.

Why I used each type of visual

I did not pick charts because they looked nice. For each one I asked: what is the reader trying to see? Here is the reason behind the five I used most.



SINGLE BIG NUMBER

Where: The first card on every page, such as Total Revenue $24.91M.

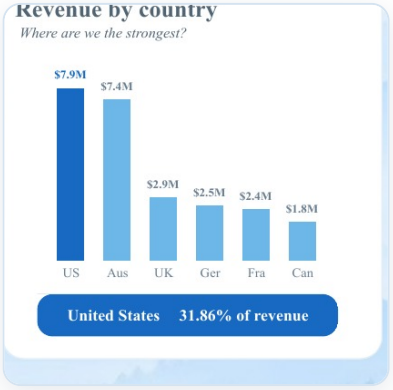
A leader wants the answer in one second. One large number says it before any chart is read. The small numbers underneath, like gross profit and orders, give support without competing. I made the first card dark blue and bigger so the eye starts there.



SPARKLINE

Where: Inside the KPI cards, such as Growth, month over month.

A sparkline is a tiny line with no axes. It shows only direction: going up, flat or down. It adds the story of time to a number without taking a whole chart of space.



COLUMN CHART

Where: Revenue by country on Overview, monthly revenue, seasonality and revenue per customer.

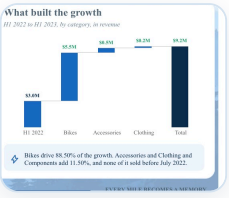
Columns are the easiest way to compare sizes side by side, because the eye reads height fast. The top column is dark blue and the others are sky blue.

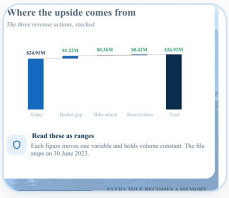


SCATTER CHART

Where: Products (revenue against margin) and Territory (basket size against growth).

A scatter shows two measures at once. A table cannot show that a product earns a lot but keeps little. The corners say Promote, Expand, Review or Reprice, so one look sorts the products into actions.





WATERFALL CHART

Where: Trends (what built the growth) and Strategy (where the upside comes from).

A waterfall shows how a total is built step by step. On Trends it starts at H1 2022 and adds bikes, accessories and clothing to reach H1 2023. On Strategy it starts at today's revenue and adds each action. The reader can see which step matters most.

How the controls connect to the charts

Everything sits on one model, so one control can drive many visuals. I kept the controls simple and put them in the same place on every page.

Control	Table and measure behind it	What it changes
Date picker (top right)	Dim_Calendar	Every visual on the page, and every KPI and caption
Sidebar buttons	Nav Page and Active Page	Page navigation, and which button looks active
Revenue / Profit / Orders	Metric Selector and Selected Metric	The Overview chart, its three side stats and its subtitle
All / Bikes / Accessories	View Toggle and Active View	The category table and the territory table
Revenue / Count	Segment View and Active Segment View	The customer segment table
Margin / Volume	Scatter Mode and Selected Scatter Mode	The product scatter chart and its Y axis
Four sliders	Slider tables and Value measures	Strategy hero number, card estimates and the upside chart

What a connected page looks like

The Overview page is the best example. The date picker filters the model. The model feeds measures. The measures feed cards, charts and captions. The Revenue, Profit and Orders buttons change one measure called Selected Metric, so one chart tells three stories without a new page.



Overview page. Cards, chart, donut, bars and tiles all read from the same model and move together when I change the date picker.

3
visual groups per page: KPI row, main chart, bottom row

1
date picker controls the whole page

4
sliders drive the Strategy numbers

11 New columns, tables and parameters

What I created that was not in the raw files

The raw files only had sales and descriptions. To answer real questions I made 17 small tables, 5 DAX columns and 6 what-if parameters. I kept each one tiny and with one job.

New tables

Table	How	Rows	What it does
Customer RFM	DAX	17,416	One row per customer with recency, frequency, spend, segment and decile
Order Basket	DAX	25,164	One row per order, with bike and accessory flags. No measure reads it since the basket fix (Appendix C).
Sales Trend & Forecast	DAX	33	30 months of actuals plus 3 forecast months with a low and high range
Nav Page	DAX	6	The six page names and their order, for the sidebar
View Toggle, Segment View	DAX	3 and 2	Labels for the toggle buttons
Customer Group, Bridge Step, Strat Upside Step	DAX	3, 4, 4	Row labels for small tables and the waterfall charts
Metric Selector, Scatter Mode	Enter data	3 and 2	Revenue, Profit, Orders and Margin, Volume buttons
4 slider tables	DAX	21 each	Numbers from 0 to 100 in steps of 5, made with GENERATESERIES
Price Adjustment %, Bike Attach Rate Uplift %	Parameter	10 and 7	Earlier what-if ideas I tested for the Strategy page

New columns

Dim_Territories[Country Short]

```
Country Short = SWITCH( Dim_Territories[Country],
    "United States", "US", "Australia", "Aus",
    "United Kingdom", "UK", "Germany", "Ger",
    "France", "Fra", "Canada", "Can",
Dim_Territories[Country] )
```

Dim_Products[Short Label] (Black, 46 becomes B46)

```
Short Label =
LEFT( Dim_Products[ProductColor], 1 ) &
Dim_Products[ProductSize]
```

Dim_Calendar[Year Label]

```
Year Label =
IF( Dim_Calendar[Year] = 2023, "2023, half year",
    FORMAT( Dim_Calendar[Year], "0" ) )
```

'Customer RFM'[Decile Label]

```
Decile Label = "Decile " & FORMAT( 'Customer
RFM'[Decile], "0" )
```

I also made Month Date in the forecast table. I made these as columns because a chart needs to put them on an axis or a label. Everything else stayed a measure.

The customer segments table

Customer RFM is my most important new table. RFM means **R**ecency (when did they last buy), **F**requency (how often) and **M**onetary (how much). I turn those three numbers into five simple groups, and then rank customers into ten equal groups called deciles.

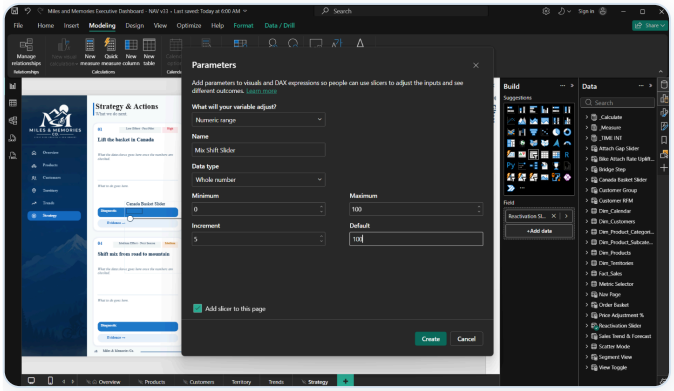
Rule, checked in this order	Segment
Last order more than 365 days ago	At risk
Last order more than 180 days ago	Lapsing
Only one order	Growing
Spent more than $3,000	High value
Everyone else	Loyal

Inside the Customer RFM table

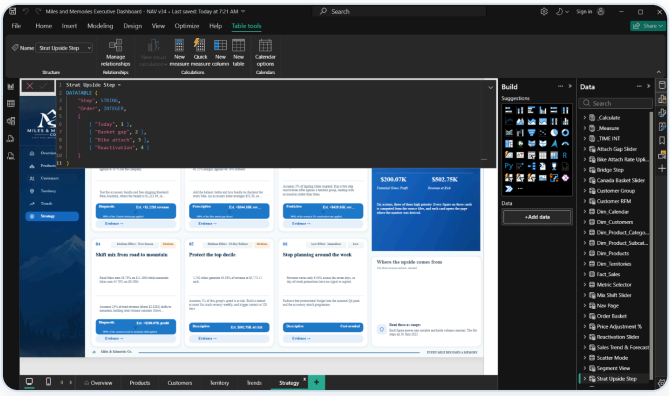
```
Segment = SWITCH( TRUE(),
    [Recency] > 365, "At risk",
    [Recency] > 180, "Lapsing",
    [Frequency] = 1, "Growing",
    [Monetary] > 3000, "High value",
    "Loyal" )
```

Parameters and what-if sliders

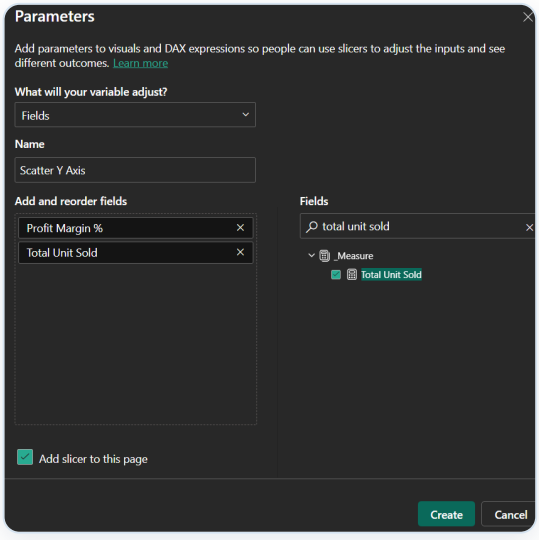
A parameter is a table that lets a person choose a value. I made **what-if** sliders for the Strategy page. For the scatter chart I first tried a **field parameter** to swap the Y axis, then settled on a small table and a measure because it gave me full control of the labels and the order.



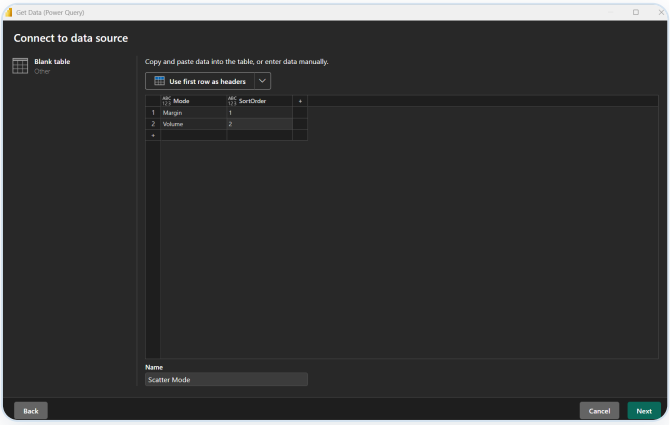
New Parameter dialog. Mix Shift Slider goes from 0 to 100 in steps of 5.



Strat Upside Step. A small DATATABLE gives the waterfall chart its four steps.



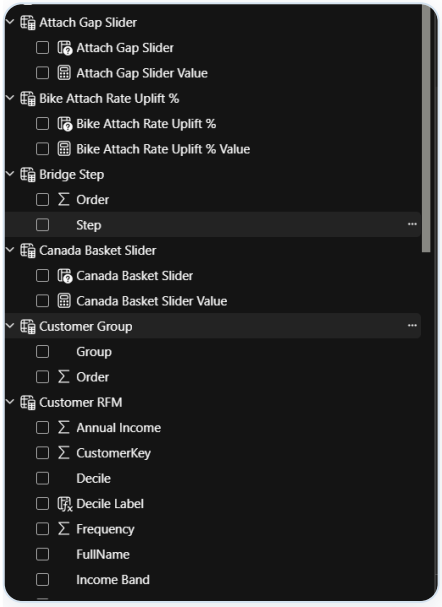
Field parameter test. My first try at swapping the scatter Y axis (Scatter Y Axis).



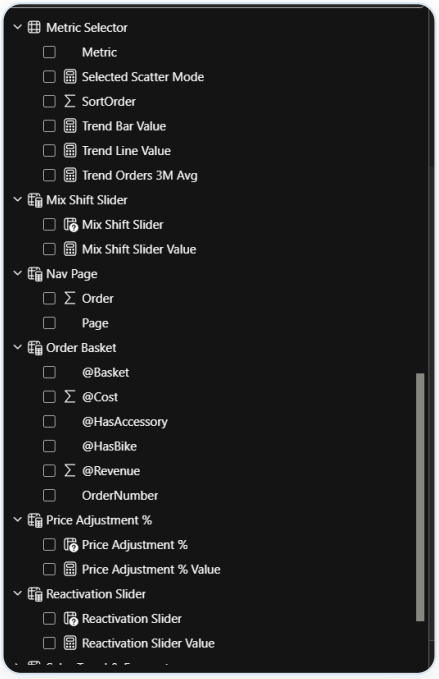
Scatter Mode. A blank table made with Enter Data holds the two button labels.

How the parameter tables look in my Data pane

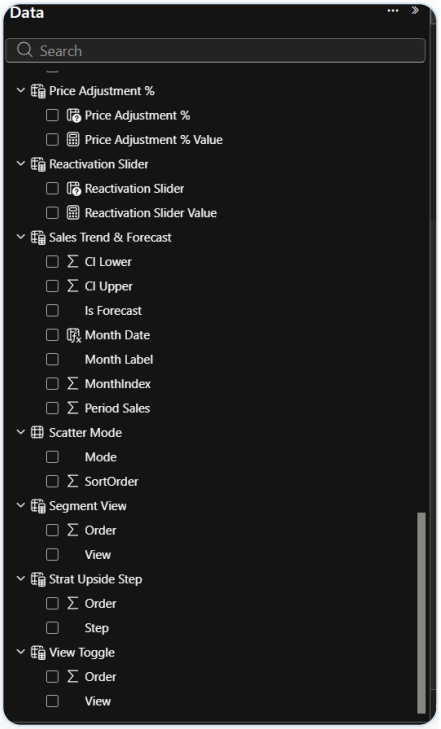
Each parameter table holds a column of choices and, when I need them, a measure that reads the choice. I name them clearly so I never wonder what a table is for.



Slider table. A column from 0 to 100 and its Value measure.



Selector table. Choices plus the measure that reads them.



Toggle table. Labels, an order, and a reader measure.

Why I made so many small tables

- **A button needs something to point at.** A toggle button writes its choice into a one-column table, and a measure reads it.
- **Sorting.** Each label table has an Order column, so buttons always appear in my order and not A to Z.
- **No filter side effects.** These tables have no relationships, so clicking a button never filters the sales by accident.

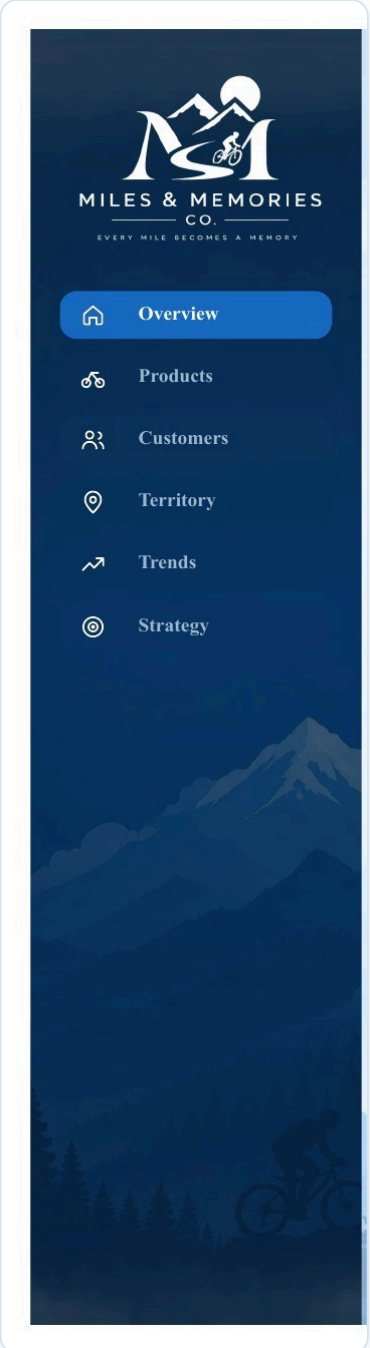
A RULE I LEARNED THE HARD WAY

A slicer or a toggle only works if the measure reads it correctly. For sliders I used MAX on the slider value, not SELECTEDVALUE, because a slider in "Before" mode is a range and SELECTEDVALUE returns blank.

12 Dashboard design decisions

Made to feel like a modern SaaS product

I wanted people to feel like they opened an app, not a report. That means a menu on the left, clean cards, one idea per card and the same layout on every page.



My sidebar.

- **Left sidebar menu.** Six buttons with icons, my logo above them. The open page is highlighted in blue.
- **Greeting and date.** "Good morning, Adventurer!" sets a friendly tone, and the date picker sits next to it.
- **KPI row first.** The first card on every page is dark blue and bigger, so the eye starts there.
- **Rounded white cards** on a soft blue sky background. One card, one idea.
- **"What this means" panel.** On the right of every page, three plain sentences and one recommendation, so no one has to guess.
- **Footer** with the company name and the brand line.

The canvas and the grid

I set the page size to **1920 x 1080** (Full HD) so it looks sharp on a big screen and scales down well. Every page uses the same four zones: sidebar, KPI row, main chart row and bottom row. Because the zones never move, people learn the layout once.

My JSON theme file

Instead of colouring each chart by hand, I wrote a **JSON theme** and imported it into Power BI. A JSON file is only a text file with settings. Mine sets the brand colours, text styles and the look of cards, slicers and tables. So every new chart I add already looks like Miles & Memories.

miles-memories-theme.json (shortened)

```
{
  "name": "Miles & Memories Co.",
  "dataColors": [
    "#1769C2", "#6DB7E8",
    "#0B2D4D", "#1E9E6B",
    "#D94F5C", "#C8842E",
    "#A8CFEF", "#526575" ],
  "background": "#FFFFFF",
  "foreground": "#0B2D4D",
  "tableAccent": "#1769C2",
  "good": "#1E9E6B",
  "bad": "#D94F5C",
  "neutral": "#C8842E",
  "textClasses": [ "title", "header",
                   "label", "callout" ],
  "visualStyles": [ "*", "card",
                    "slicer", "tableEx" ]
}
```

Colour	Where I use it
Adventure Blue	Main bars, active buttons, key cards
Sky Blue	Everything that is not the highlight
Deep Navy	Text and the biggest numbers
Green #1E9E6B	Good: growth, margin up
Red #D94F5C	Bad: decline, margin down
Amber #C8842E	Be careful: Reprice, Win back

THE RESULT

One theme file, one set of colours, and a layout that never changes. That is what makes six different pages feel like one product.

What the theme does for me

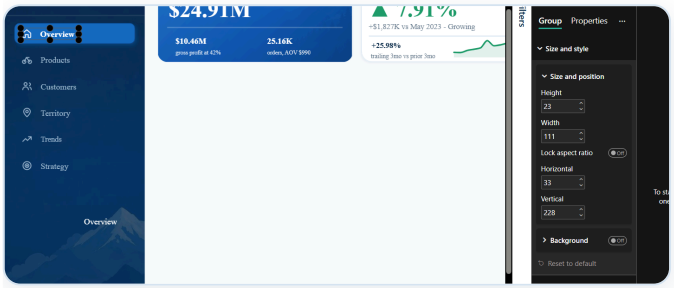
- **Saves time.** A new chart is born with the right colours, so I do not repaint it.
- **Keeps me honest.** I cannot drift into a random colour on page five.
- **Is easy to change.** If the brand blue changes, I change one line and import the file again.
- **Can be shared.** Anyone can import the same file and get the same look.

A SMALL THING ABOUT THE THEME

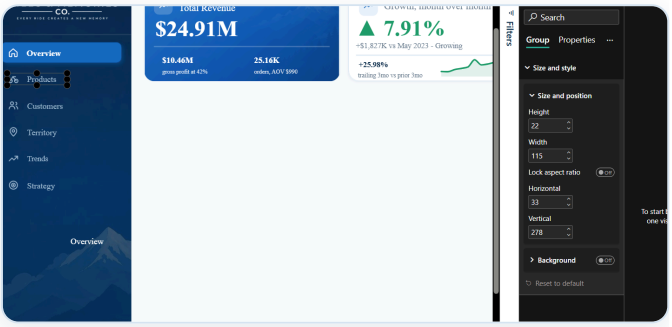
Power BI checks a theme file strictly. A single setting it does not accept can stop the whole file from loading. So I kept the file short, and I added settings one by one. The version I use has no shadow setting.

Page navigation

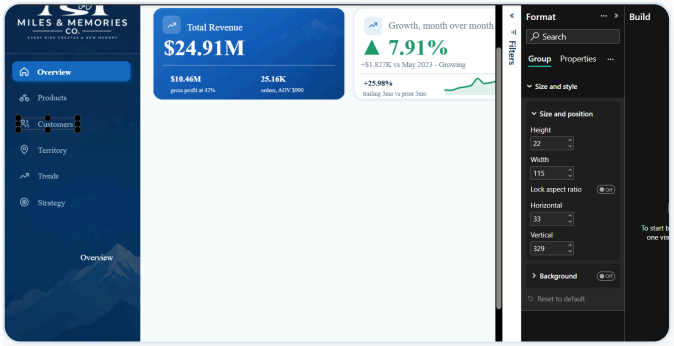
I made my own menu with buttons and bookmarks. Each of the six sidebar buttons is the same size and sits in the same position on every page. When I click one, Power BI jumps to that page. The Active Page measure makes the matching button turn blue.



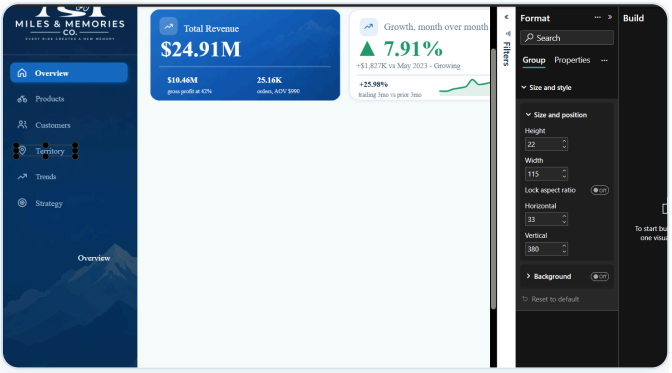
Overview button. Same width, same left edge, a different vertical position.



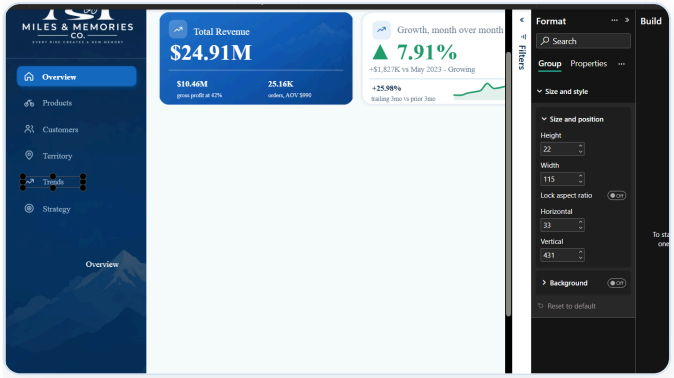
Products button. Same width, same left edge, a different vertical position.



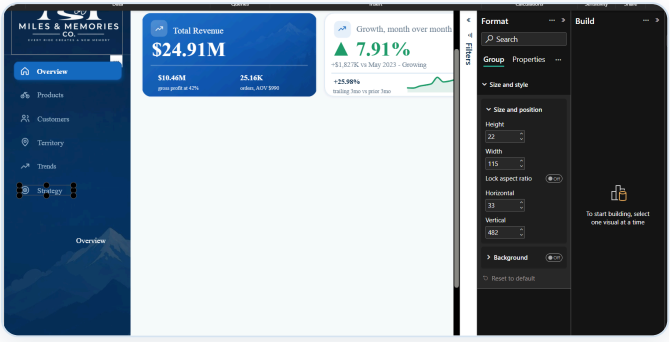
Customers button. Same width, same left edge, a different vertical position.



Territory button. Same width, same left edge, a different vertical position.



Trends button. Same width, same left edge, a different vertical position.



Strategy button. Same width, same left edge, a different vertical position.

I set the position of each button by number in the *Size and position* pane. The sidebar buttons share one width and one height, so they line up perfectly and stay the same when I copy them to the next page.

WHY NUMBERS AND NOT DRAGGING

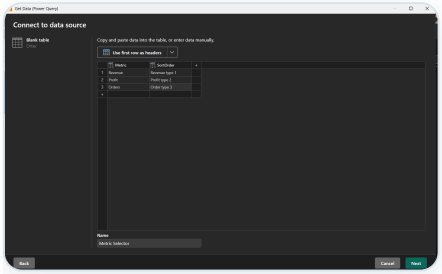
Dragging by hand is never exact. Typing the numbers means all six pages have the menu in the same place, down to the pixel. It is a small thing that makes the dashboard feel finished.

Toggle buttons

Some charts need to show different views of the same data. A **toggle** lets the viewer switch without leaving the page. Mine are on the Overview chart and the Trends line chart (Revenue, Orders), the category table (All, Bikes, Accessories), the customer segments (Revenue, Count) and the product scatter (Margin, Volume).

HOW ONE TOGGLE IS BUILT

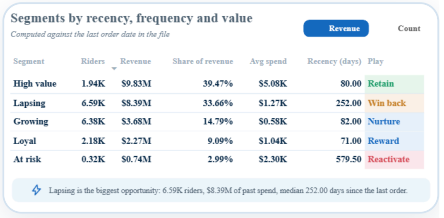
1. I made a small table with the button names, using Enter Data.
2. I added a measure that reads the choice, such as Selected Metric.
3. I made one bookmark for each button.
4. I used bookmark actions on the buttons and lined them up in the selection pane.
5. I changed the chart, card and subtitle measures to use the choice.



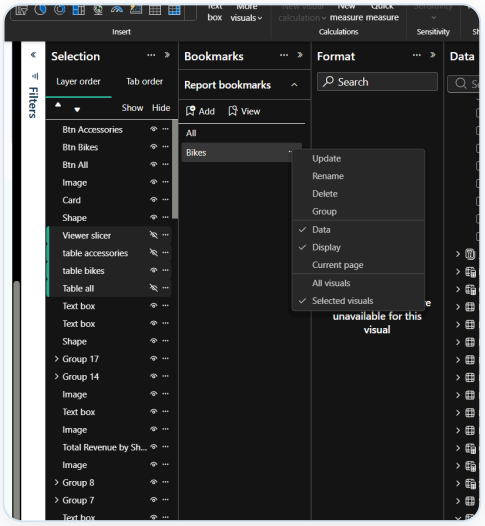
1. Metric Selector made with Enter Data.



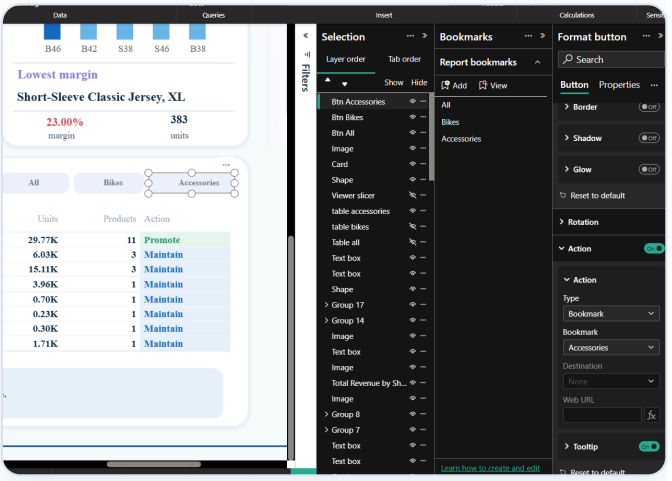
1. A table for All, Bikes, Accessories.



1. A table for Revenue and Count.



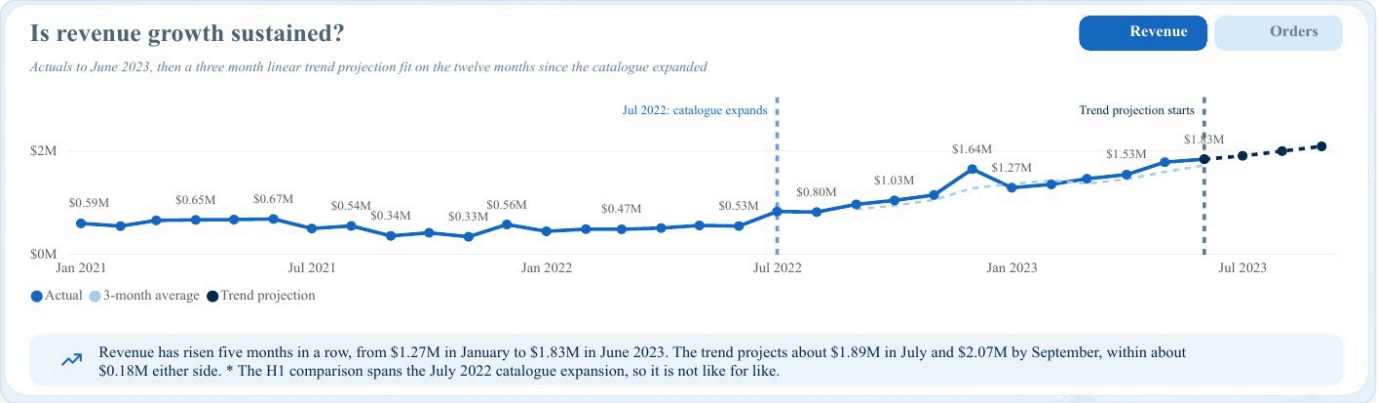
3. Selection pane and bookmarks with hidden tables.



4. The button action points to a bookmark.

The Trends toggle (added in my final review)

When I checked the dashboard one last time, I saw I had missed something. My mockup had a Revenue and an Orders button on the Trends line chart, but I had left them out of the Power BI file, so the chart only knew about revenue. A reader could not ask "are orders still climbing?". I added the two buttons to the Trends panel and made the line, the dotted average, the projection and the sentence under the chart follow the button.



The Trends panel in the final dashboard, with Revenue selected. Solid line: actual months. Dotted line: three month average. Dark dashed line: the straight line projection to September 2023. Click Orders and the line, the average, the projection and the sentence below all switch to orders.

HOW I BUILT IT

1. I copied the Metric Selector slicer from Overview into the Trends panel and hid it. It still holds the choice, so no new table was needed. In front of it I drew two buttons, Revenue and Orders, in the same pill style.
2. I wrote four measures that read the choice through Selected Metric and Trend Bar Value, the two measures the Overview toggle already uses.
3. The Revenue line uses the original chart (labels in millions). Orders needs its own chart because its labels are whole numbers, so I placed it on top.
4. I used the same bookmark pattern as the Customers page. Two bookmarks set the hidden slicer and show one chart and one pair of buttons. The selected button is solid blue and the other is pale blue, and the pairs sit in exactly the same place.
5. I pointed the sentence under the chart to the new Trend Metric Callout measure, and fixed the subtitle so it says twelve months, as the mockup does.

A CORRECTION I MADE

The subtitle used to say the projection was fitted on the "thirty months" since the catalogue expanded. That was wrong. The line is fitted on **twelve months, July 2022 to June 2023**. The chart still shows all 30 months of actuals, but the fit only uses the last 12, because the catalogue changed in July 2022 and the earlier months are a different business. I corrected the measure and every place in this case study that repeated it.

WHAT WENT WRONG FIRST

My first version hid the Orders chart with a filter. The chart went empty, but it still drew its two reference lines on top of the Revenue chart, so "Trend projection starts" appeared twice. A hidden chart has to be hidden with a bookmark, not emptied with a filter.

The four measures behind the Trends toggle

Measure	Table	What it does
Trend Metric Actual	_TIME INT	Solid line. For Revenue it uses the old Actual Sales. For Orders it reads Trend Bar Value for that month.
Trend Metric 3M Avg	_TIME INT	Dotted line. The average of the last three months of the chosen metric, from September 2022.
Trend Metric Projection	_TIME INT	Dark dashed line. A straight line fitted on July 2022 to June 2023, then read at July, August and September 2023.
Trend Metric Callout	_Measure	The sentence under the chart. It finds the winning streak and writes the projection in words, in dollars or in orders.

All four start with the same safety rule: **if the button is Revenue, use the measure I already trusted**. That way the Revenue view cannot change by accident. Orders follows the same method on different numbers.

Trend Metric Actual

```
Trend Metric Actual =
VAR d = MAX ( 'Sales Trend & Forecast'[Month Date] )
VAR i = MAX ( 'Sales Trend & Forecast'[MonthIndex] )
RETURN
    IF (
        [Selected Metric] = "Revenue",
        [Actual Sales],
        IF (
            i <= 12,
            CALCULATE (
                [Trend Bar Value],
                ALL ( Dim_Calendar ),
                Dim_Calendar[Start of Month] = d
            )
        )
    )
```

- **MAX of Month Date** reads which month the chart is drawing.
- **[Selected Metric]** reads the button. Revenue keeps the old measure.
- **ALL(Dim_Calendar) plus the Start of Month test** is needed because the trend table has no relationship to the calendar, so I point to the month myself.
- **i <= 12** stops the solid line at June 2023, so it does not run into the forecast months.

WHAT EACH BUTTON SAYS (CHECKED OUTSIDE POWER BI WITH THE SAME METHOD)

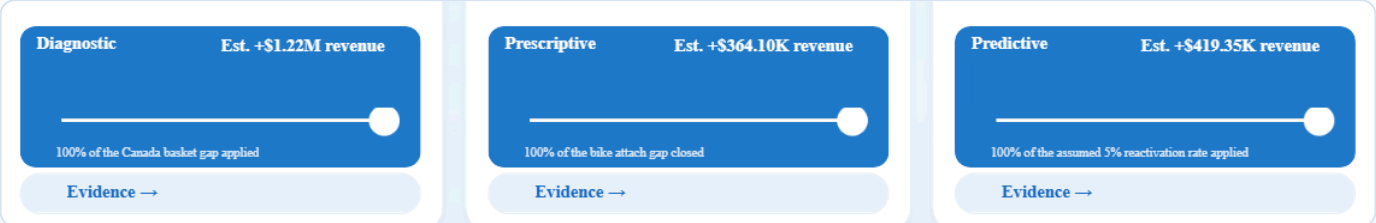
Button	June 2023	Projected July	Projected September	The sentence says
Revenue	$1.83M	$1.89M	$2.07M	Up five months in a row, from $1.27M in January.
Orders	2,146	2,378	2,575	Fell in June, so the run of gains has ended.

WHAT ORDERS ADDS

This is the reason for the toggle. Revenue grew 3.31% in June, but orders slipped 0.88% (2,165 in May to 2,146 in June). The extra revenue came from bigger baskets, with average order value up 4.22% to $851.35. Revenue alone hides that the number of orders paused. The projection still points up for both, but the Orders view is the one to watch next month.

Scroll bars (sliders) on the Strategy page

On Strategy, I wanted people to **play** with the numbers. Each of the first three cards, and the road to mountain card, has a slider like a scroll bar. When I drag it, the estimate on the card, the big number on the right and the chart below all change.



The slider pill on a card. The label says what the slider is, such as "Diagnostic", and the pill moves from 0 to 100.



The card header. Effort, priority and the estimate sit together.

4
sliders, 0 to 100% in steps of 5

3
revenue gaps combined into the hero number

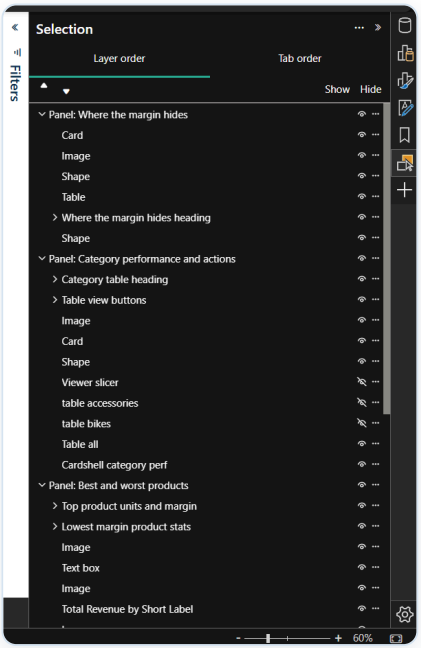
$2.01M
potential revenue at the default settings

WHAT A SLIDER DOES

The Canada slider decides how much of the Canada basket gap to count. At 100% the whole gap is counted. At 50% only half is. The estimate is simply the gap times the slider divided by 100. This lets a leader be careful or hopeful, and see the effect right away.

Organising the Selection pane

A page like Overview has about 60 objects. Without names they all say "Card", "Shape", "Image". I renamed and grouped everything so I can find a visual in seconds, and so nothing moves by accident when I click.



Group names. "Panel: Where the margin hides", "Panel: Best and worst products" and more.

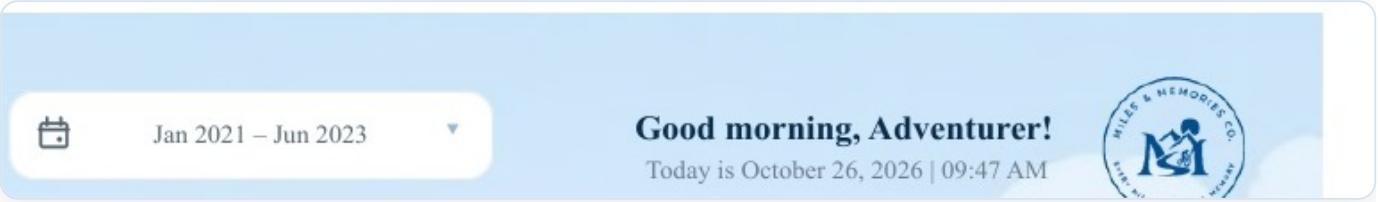


The Selection pane in the final file on the Overview page.

- **Grouped by purpose**: Sidebar navigation, Page title, Greeting and date picker, KPI cards, Panels, What this means rail, Footer.
- **Named like a sentence**, such as "Card 03: Reactivate the lapsing segment".
- **Groups keep their shape**, so moving a group moves all of its shapes, charts, icons and text together.

Greeting, date and the link to the plan

When someone opens any page, the top right says hello by itself. It says Good morning, Good afternoon or Good evening, then "Adventurer", and under it today's date and the time. I did not type any of this. Two measures work it out every time the page opens, so it is always right.



The top bar. My date picker, the greeting, today's date and time, and the round Miles & Memories badge. It is the same on all six pages.

Greeting Text

```
VAR _Hour = HOUR(NOW())
RETURN
SWITCH(
    TRUE(),
    _Hour < 12, "Good morning, Adventurer!",
    _Hour < 18, "Good afternoon, Adventurer!",
    "Good evening, Adventurer!"
)
```

Greeting Date

```
"Today is " & FORMAT(TODAY(), "mmmm d, yyyy") &
"   |   " & FORMAT(NOW(), "hh:mm AM/PM"
)
```

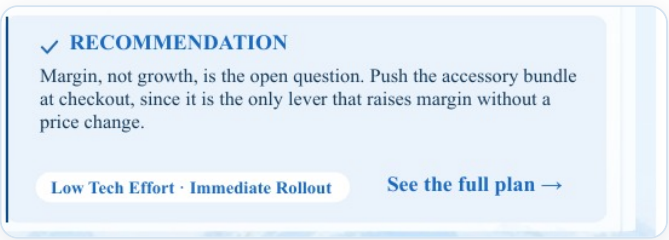
- **Morning** is before 12 noon, **afternoon** is from 12 noon to 5:59 pm, and **evening** is 6 pm and later.
- **HOUR(NOW())** reads the clock. **SWITCH** picks the right words.
- **FORMAT** turns the date and time into plain words, like "October 8, 2026 | 10:19 AM".

ONE THING I LEARNED

Power BI Desktop uses the clock on my computer. After I publish, the Power BI Service uses UTC time, which is 8 hours behind the Philippines. If I share the report online, the greeting and the time can be a few hours off. To fix it, I add 8 hours to NOW() inside both measures.

"See the full plan" button

Five pages have a "What this means" panel on the right with three plain sentences and one recommendation. At the bottom of the recommendation there is a button that says **See the full plan**. When someone clicks it, they go straight to the Strategy page, where the plan is shown with numbers.



The recommendation box. The grey tag shows the effort and the timing, and the link opens Strategy.

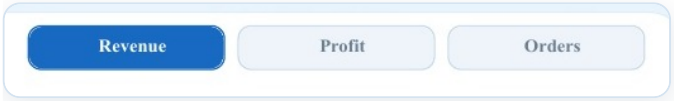
- It is a **button with a Page navigation action**. Its destination is the Strategy page.
- It is on **Overview, Products, Customers, Territory and Trends**. Strategy is the place it points to, so it does not need one.
- A reader goes from "what happened" to "what do we do" in one click.

The filters and slicers I used

I kept the filters few and easy to see. Every page has the same date picker in the same spot. Other pages add only the one or two switches that matter for that page. Nothing is hidden in a pane.

Page	Filter	What it does	Where it comes from
All six pages	Date picker (month list)	Opens a month list. Every card, chart and table on the page recalculates.	`Dim_Calendar, Start of Month`
Overview	Revenue, Profit, Orders	Changes what the growth chart shows.	`Metric Selector, Metric`
Trends	Revenue, Orders	Changes the line chart, its projection and the sentence under it.	`Metric Selector, Metric`
Products	Margin or Volume	Changes the scatter chart between margin and units.	`Scatter Mode, Mode`
Products	All, Bikes, Accessories	Changes what the category table shows.	`View Toggle, View`
Customers	Revenue or Count	Switches the segment table between money and riders.	`Segment View, View`
Territory	All, Bikes, Accessories	Changes what the country table shows.	`View Toggle, View`
Strategy	Four sliders, 0 to 100% in steps of 5	Lets people change how much of each gap to count.	`Reactivation, Canada Basket, Attach Gap and Mix Shift sliders`

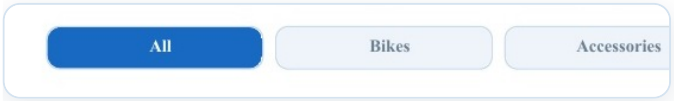
HOW THEY LOOK ON THE DASHBOARD



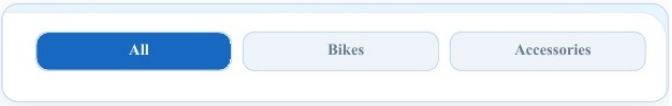
Overview. Revenue, Profit and Orders.



Products. Margin or Volume.



Products. All, Bikes or Accessories.



Territory. The same three choices.



Customers. Revenue or Count.



Strategy. Sliders for the three revenue cards.

HOW THEY WORK

The date picker filters the model through the calendar. The other switches (Metric, Mode, View and the sliders) come from small tables that have no relationships. A measure reads the choice with SELECTEDVALUE or MAX and decides what to show. This is why clicking a button never filters the rest of the page by accident.

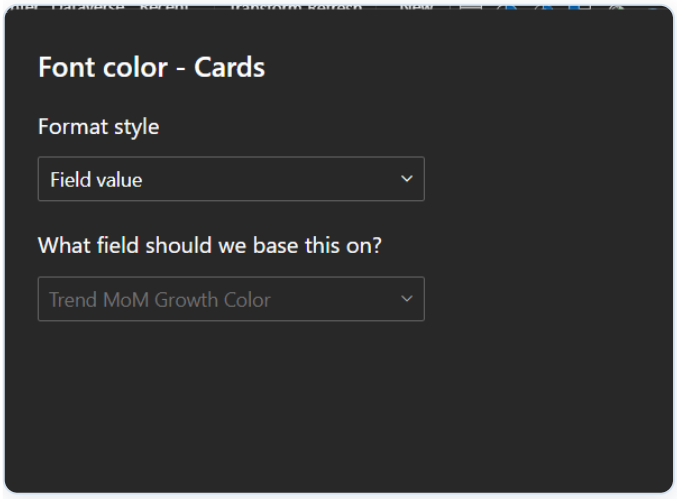
13 Visualisation choices

Colour that talks, lines that explain

I never use colour only to make a chart pretty. A colour has to say something. I use measures and rules to pick the colour, so it changes by itself when the data changes.

Data labels with conditional colours

The month over month chart shows a green label when the month grew and a red label when it fell. The colour comes from a measure. In the format pane I chose *Font colour → Format style: Field value* and pointed it to my colour measure.



Field value. The label colour is read from a measure, here Trend MoM Growth Color.

MoM Label Color

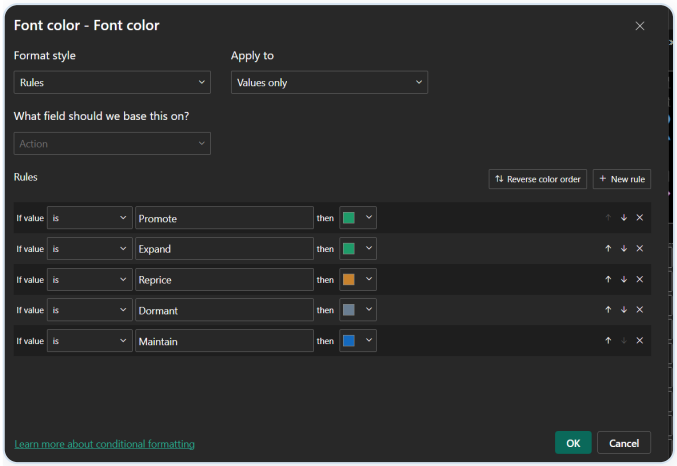
```
IF ( [MoM Chart Growth %] < 0, "#D94F5C", "#1E9E6B" )
```

MoM Fill Color (the bar itself)

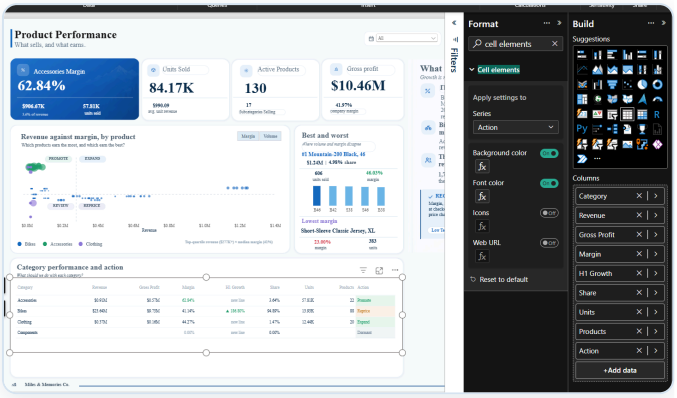
```
IF ( [MoM Chart Growth %] < 0, "#CFE3F5", "#1769C2" )
```

Colour by rules

In tables I used two ways. The first is **Rules**, which needs no DAX. In the category table, the word in the Action column decides the colour: Promote and Expand are green, Reprice is amber, Dormant is grey and Maintain is blue. The second is **Field value**, when the rule needs a calculation.



Rules. If the value is "Promote", the text is green. Each Action word has its own colour.



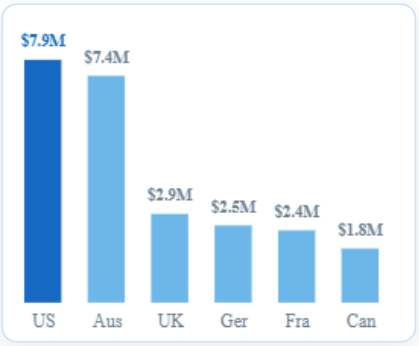
Cell elements. I switch on Background colour and Font colour for the Action column.

RULES OR DAX?

I used **rules** when the colour depends on a fixed word or one simple cut, like above or below zero. I used **DAX** when the colour depends on other things, such as "is this the best month in the months I can see?". DAX is more work, but it never goes out of date.

Conditional chart colours and the track gap

On the country chart the biggest market is dark blue and the others are sky blue. This is not painted by hand. A measure checks which bar is the highest and gives it the dark colour, so if the filter changes, the dark bar moves with it.



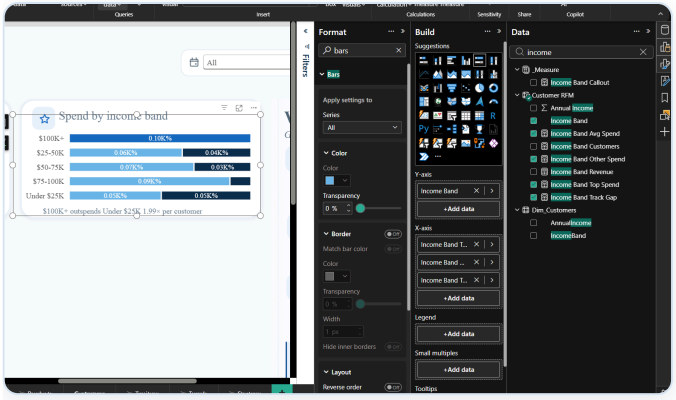
Result. The top country is Adventure Blue and the rest are Sky Blue.

Seasonality Bar Color

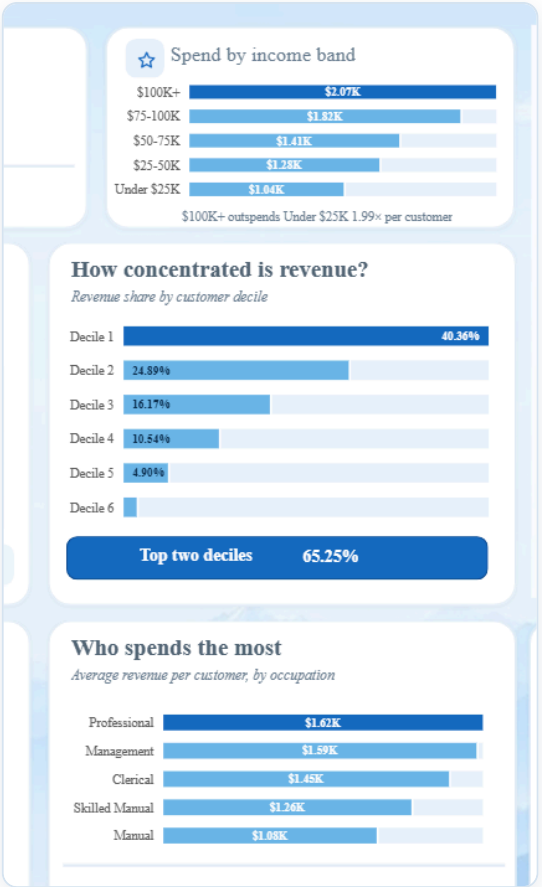
```
VAR Cur = [Total Revenue]
VAR Best = MAXX ( ALLSELECTED (
Dim_Calendar[MonthName_Short] ), [Total Revenue] )
RETURN
    IF ( Cur = Best, "#1769C2", "#6DB7E8" )
```

THE TRACK GAP

Some bars sit on a pale **track** that shows how far a bar could go. I made the track with a measure called **Track Gap**. It is the biggest value minus this bar's value. I stack it on top of the real bar in a light colour, so every row reaches the same total length. The real bars then look like progress bars.



Formatting the bars. The Data pane shows my Income Band measures, including Track Gap, Top Spend and Other Spend.



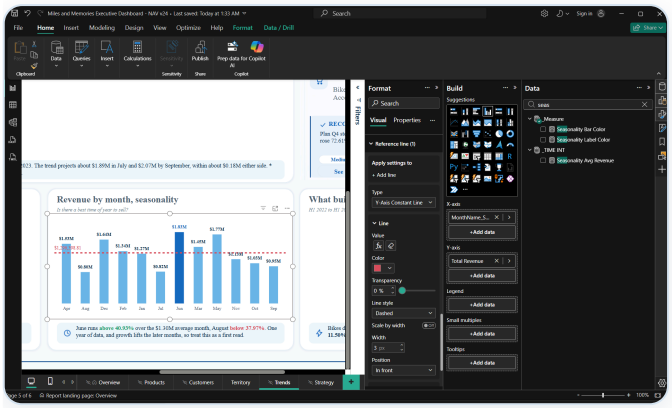
Result on the Customers page. Decile and income bars with their light tracks.

Income Band Track Gap

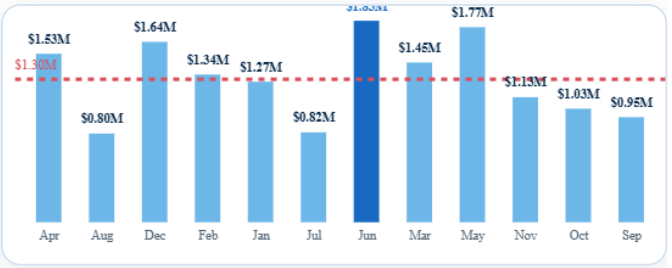
```
MAXX ( ALLSELECTED ( 'Customer RFM'[Income Band] ),
[Income Band Avg Spend] )
    - [Income Band Avg Spend]
```

Reference lines

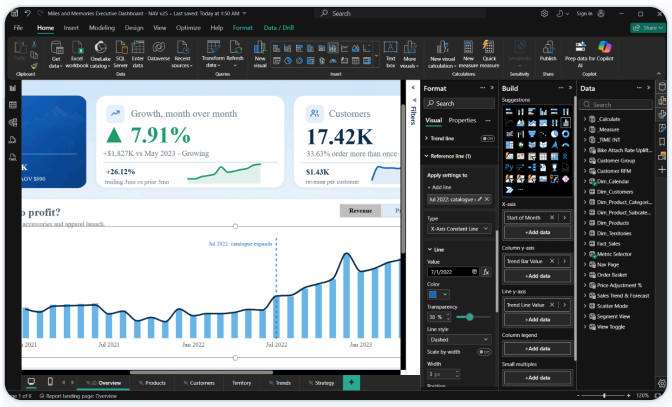
A reference line is a line across a chart at a value that matters, so a viewer sees at once what is above and below. I used them in three ways: an **average line** on the seasonality chart, a **vertical line** on July 2022 when the catalogue expanded, and a **start of projection** line on the trend.



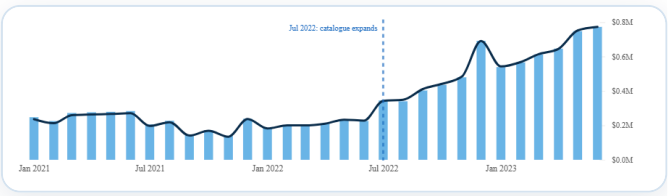
Y-axis constant line. Set in the Analytics pane: dashed, red, placed in front.



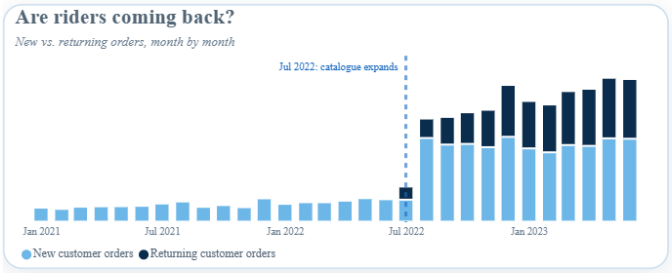
Result. Months above the dashed line beat the average, months below do not.



X-axis constant line. A date value, 1 July 2022, with the label "Jul 2022: catalogue expands".



Result. The vertical line shows where growth changed.



Same line on the customers chart. Returning orders only start after the line.



Trends chart. One line for the catalogue, one for where the projection starts.

14 Key insights

What the data told me, page by page

This is the most important part. Here are the six pages as they look in the final dashboard (each one is a full 1920 x 1080, 16:9 page), and what I learned from each one. The next pages go through them one by one.

The big picture in five sentences

- **Growth is real.** Revenue went from about $0.53M a month before July 2022 to $1.83M in June 2023.
- **The catalogue caused it.** Accessories and clothing only started in July 2022, and bikes still brought 88.50% of the extra revenue.
- **Margin is the open question.** Margin slipped 0.13 points even as sales grew.
- **Money is concentrated.** Bikes are 94.89% of revenue and the top 10% of customers bring in 40.36%.
- **The next dollars are in baskets and returning riders.** Canada's baskets are small, half of bike orders have no accessory, and 6.59K riders have gone quiet.

$2.01M
revenue in the three revenue gaps

$200.07K
extra gross profit from moving road to mountain

$502.75K
revenue at risk in the top decile

211.07%
H1 2023 against H1 2022

The six pages at a glance

Page	Headline I wrote	Main action
Overview	Growth is real. Margin is the open question.	Push the accessory bundle at checkout
Products	Revenue and margin come from different places.	Bundle a helmet, a bottle and tires with every bike
Customers	A small share of riders carries the business.	Run a two step reactivation offer
Territory	Basket size, not margin, separates the markets.	Pilot a bike-led offer in Canada
Trends	Growth has held since July 2022.	Plan Q4 stock and staffing for the December spike
Strategy	Six actions, each with a value.	Start with Canada, attach and reactivation

PAGE 1 Overview

Is growth translating into profit?



WHAT I FOUND

The catalogue changed the business

Before July 2022 only bikes were sold at about $533.82K a month. By June 2023 accessories and apparel lifted it to $1.83M.

Bikes carry revenue, accessories carry margin

Bikes are 94.89% of revenue at 41.14% margin. Accessories earn 62.84% but are only 3.64% of revenue.

Customers are concentrated

The top tenth of customers (1,742 riders) bring in 40.36% of revenue at $5.77K each. Losing 5% of them costs about $502.75K.

WHY I BUILT THE PAGE THIS WAY

A column chart with a smoothed line answers "is growth real?" without one big month fooling anyone. The basket effect tiles sit next to it because margin, not sales, is the open question. The three buttons let one chart show revenue, profit or orders.

Measures behind it: Trailing 3M Revenue, MoM Chart Growth %, Bike Attach Rate %, Selected Metric

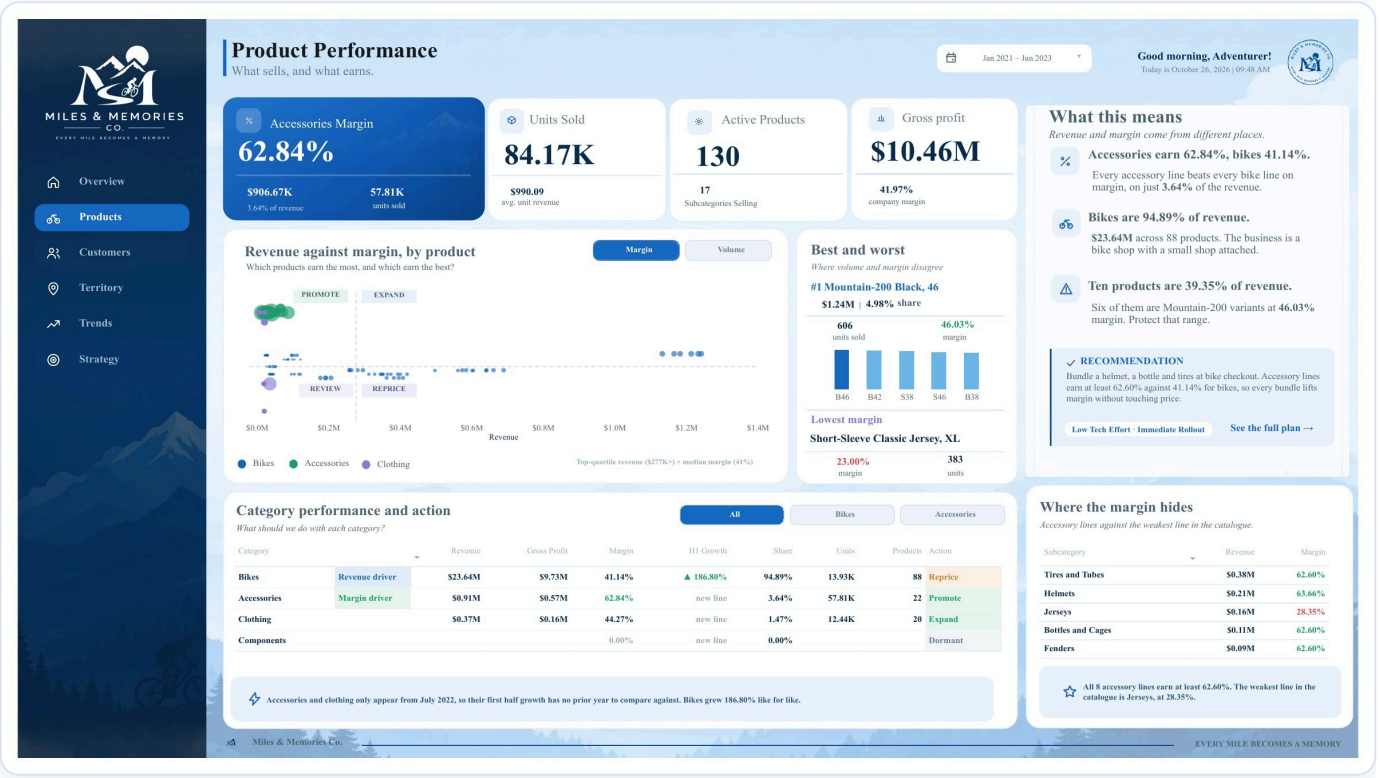
WHAT I WOULD DO

Push the accessory bundle at checkout. It is the only lever that raises margin without changing the price. Orders with an accessory earn a 42.21% margin against 40.79% without, and 51.04% of bike orders already have one.

PAGE 2

Products

What sells, and what earns?



WHAT I FOUND

Every accessory line beats every bike line

Accessory lines earn at least 62.60% margin, while bikes earn 41.14%. The weakest line in the whole catalogue is Jerseys at 28.35%.

Ten products are 39.35% of revenue

Six of them are Mountain-200 variants at a 46.03% margin. This is the range to protect.

Category calls are clear

Bikes: Reprice. Accessories: Promote. Clothing: Expand. Components: Dormant, with no sales.

WHY I BUILT THE PAGE THIS WAY

The scatter chart puts revenue on one side and margin on the other, so the four corners say Promote, Expand, Review or Reprice at a glance. The table beside it turns the same idea into one call for each category.

Measures behind it: Category Action, Scatter Dot Color, Selected Scatter Mode, Active View

WHAT I WOULD DO

Bundle a helmet, a bottle and tires at bike checkout. Accessory lines earn at least 62.60% against 41.14% for bikes, so every bundle lifts margin without touching price.

PAGE 3

Customers

Who rides with us, and who comes back?



WHAT I FOUND

Revenue is heavily concentrated

The top two deciles are 65.25% of revenue. Deciles six to ten together are only 3.14%.

Only 33.63% order twice

The average rider has placed 1.44 orders. Returning orders are zero before July 2022 and reach 42.45% of monthly orders by June 2023.

Lapsing is the biggest opportunity

6.59K riders hold $8.39M of past spend, and a median of 252 days has passed since they last ordered.

WHY I BUILT THE PAGE THIS WAY

Tables give the exact numbers for each customer group. Bars with a light track show how fast revenue falls from the top decile to the rest. The stacked columns show whether new riders turn into returning riders.

Measures behind it: Top Decile Share %, Repeat Customer Rate %, Income Band Track Gap, Decile Track Gap

WHAT I WOULD DO

Run a two step reactivation offer for the 6.59K lapsing riders, and measure it against a holdout group before rolling it out wider.

PAGE 4 Territory

Where are the miles being made?



WHAT I FOUND

Two markets are 61.63% of revenue

The United States ($7.94M) and Australia ($7.42M). The next four markets together are 38.37%.

Canada sells often and small

3.02K orders at $585.07 each, against $1.22K in Australia. Only 28.37% of Canadian orders include a bike, against 55.35% overall.

Margin is flat across markets

It only ranges from 41.49% to 42.83%. What separates the markets is basket size, not margin.

WHY I BUILT THE PAGE THIS WAY

One table compares countries on size, basket and growth together. The bubble chart then shows why size alone is not the story, because a market can be big but have small baskets. The line on it marks the company average.

Measures behind it: Company AOV, Strat Weak Market, Fastest Growth Color, Active View

WHAT I WOULD DO

Pilot a bike-led offer in Canada, where a bike order is worth $1.88K but only 28.37% of orders include one. Measure it against a holdout.

PAGE 5

Trends

Where is the road heading?



WHAT I FOUND

Growth started in July 2022

Monthly revenue went from $0.53M in June 2022 to $0.82M in July. H1 2023 is 3.11 times H1 2022 at $9.19M.

Growth is still climbing

Five straight monthly gains, from $1.27M in January to $1.83M in June 2023. The trend points to about $1.89M in July and $2.07M by September.

Seasons matter

June is the best month at 40.93% above average. August is the weakest at 37.97% below. January is the only real fall, down 22.07%.

WHY I BUILT THE PAGE THIS WAY

A line chart with a dashed projection answers "where are we heading?". The growth and seasonality columns answer "when does it slow down?". The waterfall answers "what built the growth?". Colours say good or bad without a legend. The line chart has Revenue and Orders buttons.

Measures behind it: Trend Metric Actual, Trend Metric Projection, MoM Label Color, Seasonality Bar Color

WHAT I WOULD DO

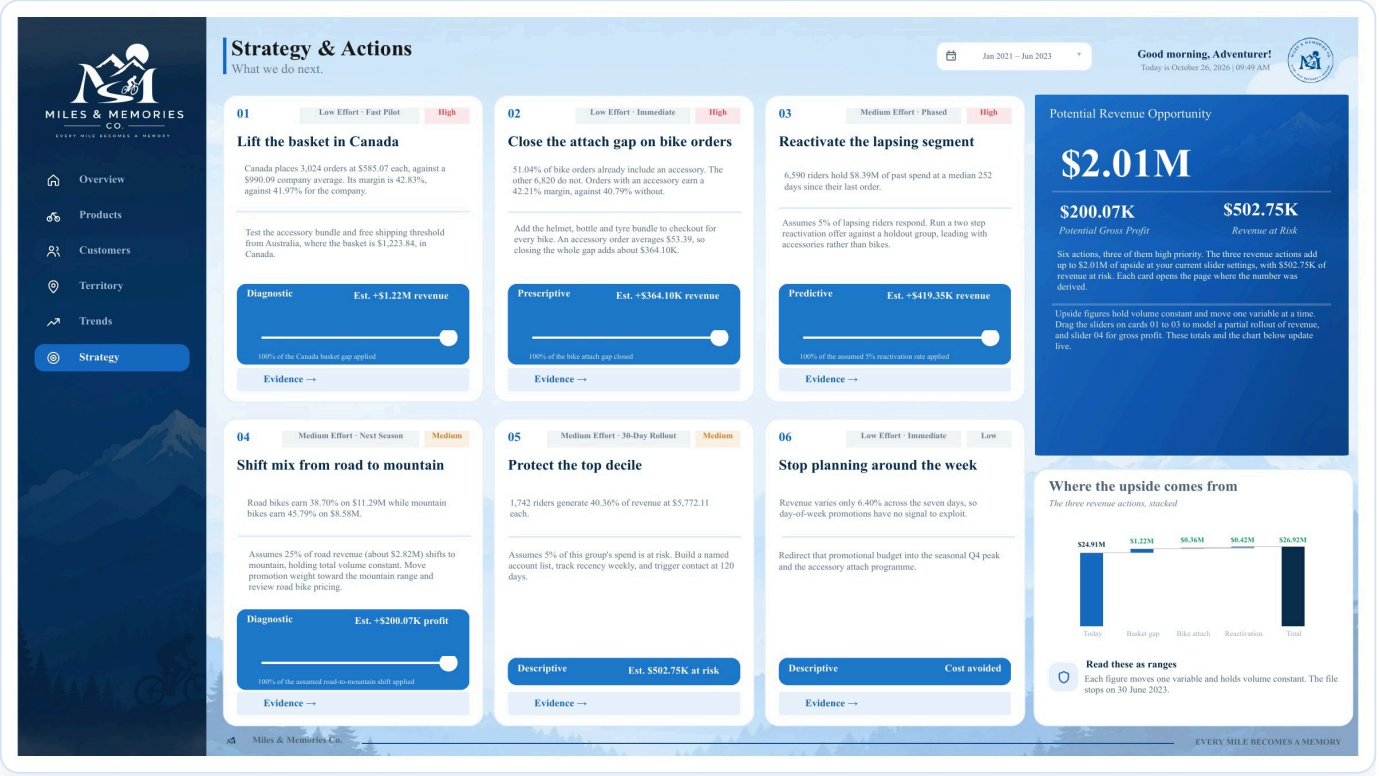
Plan Q4 stock and staffing for a December spike. December revenue rose 72.61% in 2021 and 44.22% in 2022 on the month before.

The projection is a straight line fitted to the 12 months since the catalogue expanded (July 2022 to June 2023), with an 80% range of about $0.18M either side. It is a first read, not a promise.

PAGE 6

Strategy

What do we do next?



WHAT I FOUND

Three revenue gaps add up to $2.01M

Canada basket $1.22M, bike attach $364.10K and reactivation $419.35K, at the default slider settings.

Two more actions protect profit

Road to mountain adds about $200.07K of gross profit. Protecting the top decile guards $502.75K.

One thing to stop

Revenue varies only 6.40% across the seven days, so day of week promotions have no signal to use.

WHY I BUILT THE PAGE THIS WAY

Each card holds one action with its evidence, its effort and a slider. The hero number and the chart add the cards up, so the leader sees the total and the pieces at the same time. This is where the earlier pages become a plan.

Measures behind it: Strat Basket Gap $, Strat Attach Gap $, Strat Reactivation $, Strat Revenue Upside

WHAT I WOULD DO

Start with the two low effort, high priority actions: lift the Canada basket and close the attach gap. Then add the phased reactivation offer.

Each figure moves one variable and holds volume constant. The file stops on 30 June 2023, so please read these as ranges.

The four core analyses

Descriptive, diagnostic, predictive and prescriptive

Good analysis answers four questions in order. What happened? Why did it happen? What will happen next? And what should we do? My dashboard follows this path from the first page to the last.

OVERVIEW, PRODUCTS, TERRITORY

Descriptive

What happened?

Totals, rankings and shares: $24.91M revenue, bikes 94.89%, United States as the biggest market.

Measures: Total Revenue, Total Profit, Total Orders, Profit Margin %, category and country totals

OVERVIEW, PRODUCTS, CUSTOMERS, TERRITORY

Diagnostic

Why did it happen?

Margin sits in accessories, revenue sits in bikes. Canada has small baskets. The catalogue change explains the jump in July 2022.

Measures: Basket effect, RFM segments, Track Gap, Bike Attach Rate %, MoM Chart Growth %

TRENDS, STRATEGY CARD 03

Predictive

What happens next?

A straight line fitted to the 12 months since the catalogue expanded projects July to September 2023 with an 80% range.

Measures: Sales Trend & Forecast table, Trend Metric Projection, Strat Reactivation $

STRATEGY

Prescriptive

What should we do?

Six actions with an effort level, a priority, and a value that moves with each slider.

Measures: Strat Basket Gap $, Strat Attach Gap $, Strat Revenue Upside, Strat Upside Value

How they fit together

1. Describe

I show the numbers first, so everyone starts from the same facts.

2. Diagnose

I explain the number. A fact without a reason cannot be acted on.

3. Predict

I look forward, carefully and with a range, not one number.

4. Prescribe

I turn the story into actions and put a value on each.

WHERE YOU SEE IT ON THE DASHBOARD

The Strategy page cards are tagged with the kind of analysis behind them: **Diagnostic** for the Canada basket and the road to mountain shift, **Prescriptive** for the attach gap, **Predictive** for the lapsing riders, and **Descriptive** for the top decile and the day of the week. The tag tells a reader how much to trust the number and why.

15 Business recommendations

Six actions, ranked, with the number behind each

A finding is only useful if someone can act on it. For each action I wrote the evidence, the value, how hard it is and how to test it. These are estimates that hold volume constant and move one variable at a time.

#	Action	Evidence from the dashboard	Estimated value	Effort	Priority
1	Lift the basket in Canada	Canada places 3,024 orders at $585.07, against a $990.09 company average.	+$1.22M revenue	Low effort, fast pilot	High
2	Close the attach gap on bike orders	51.04% of bike orders have an accessory. 6,820 do not.	+$364.10K revenue	Low effort, immediate	High
3	Reactivate the lapsing segment	6,590 riders hold $8.39M of past spend, 252 days since the last order.	+$419.35K revenue	Medium effort, phased	High
4	Shift mix from road to mountain	Road bikes earn 38.70% margin and mountain bikes 45.79%.	+$200.07K profit	Medium effort, next season	Medium
5	Protect the top decile	1,742 riders bring in 40.36% of revenue at $5.77K each.	$502.75K at risk	Medium effort, 30 day rollout	Medium
6	Stop planning around the week	Revenue varies only 6.40% across the seven days.	Cost avoided	Low effort, immediate	Low

How I would roll them out

QUICK WINS

First 30 days

Add the helmet, bottle and tire bundle to bike checkout. Start the Canada pilot with a free shipping threshold taken from Australia. Draft the named list of top decile riders.

TEST AND LEARN

Days 31 to 60

Run the two step offer for lapsing riders against a holdout group. Contact top decile riders when they reach 120 days without an order.

PLAN THE SEASON

Days 61 to 90

Plan Q4 stock and staffing for the December spike. Move promotion budget away from day of week offers. Review road bike prices before next season.

How I would know it worked

- **Canada:** average order value moves up from $585.07 toward the $990.09 company average, and the share of orders with a bike rises from 28.37%.
- **Attach:** the bike attach rate rises from 51.04%, and the margin of bike orders rises with it.
- **Reactivation:** responders in the test group order more than the holdout group.
- **Top decile:** fewer riders pass 120 days without an order.

BE HONEST ABOUT THE NUMBERS

The data stops on 30 June 2023 and the response rates are my assumptions, such as 5% of lapsing riders responding. So I show the results as ranges and ask for a pilot before a full rollout. The sliders on the Strategy page let any leader change those assumptions.

Key takeaways, summary and conclusions

The short version, for anyone who reads only one page

Key takeaways

- **Growth is real.** Monthly revenue went from $0.53M in June 2022 to $1.83M in June 2023.
- **The catalogue change caused it.** Accessories and clothing started in July 2022.
- **Margin is the open question.** It slipped 0.13 points while sales climbed.
- **Revenue is concentrated.** Bikes are 94.89% and the top 10% of customers are 40.36%.
- **Accessories carry the margin.** 62.84% margin on 3.64% of revenue.
- **Few riders come back.** Only 33.63% order twice, and 6.59K riders have gone quiet.
- **Canada sells often but small.** $585.07 per order against $990.09 for the company.
- **Momentum is climbing.** Five straight monthly gains and a projection of $2.07M by September 2023.

Summary

Miles & Memories Co. earned **$24.91M in revenue and $10.46M in gross profit** from 25.16K orders and 17.42K customers. After accessories and clothing joined a bike-only catalogue in July 2022, monthly revenue more than tripled. But the business is concentrated in one category, a small group of customers and two markets, and its margin has not grown with its sales.

I turned the findings into **six actions**: about **$2.01M** of extra revenue from three gaps (Canada baskets, bike attach and lapsing riders), about **$200.07K** of extra gross profit from a better mix, and **$502.75K** of revenue to protect in the top customer group. All of it sits in one model, so every number can be traced back to its source.

Conclusions

#	Conclusion	Supported by
1	**Growth is real and was caused by the catalogue change in July 2022.**	Revenue up from $0.53M to $1.83M a month; bikes bring 88.50% of the extra revenue.
2	**Sales growth has not yet become better profit.**	Margin down 0.13 points over three months; accessories are only 3.64% of revenue.
3	**The business depends on a few customers and one category.**	Top 10% of customers = 40.36% of revenue; bikes = 94.89%.
4	**Basket size, not margin, separates the markets.**	Margin only ranges 41.49% to 42.83%; Canada basket $585.07 against $1.22K in Australia.
5	**The next gains are in baskets and returning riders.**	48.96% of bike orders have no accessory; 6.59K lapsing riders hold $8.39M of past spend.
6	**Growth is expected to continue, with a seasonal peak.**	Trend of $1.89M in July and $2.07M in September; December +72.61% and +44.22%.

16 Technical challenges

What went wrong, what I did and what I learned

This project was not smooth. Here are the problems that cost me the most time. I list them because fixing them taught me more than the parts that worked the first time.

1	**My slider numbers kept showing**	**PROBLEM** The slicer I used for each slider showed a number box I did not want.	**WHAT I DID** I made a cover strip, a shape in the page colour, and made it tall enough to hide the box and the borders.	**WHAT I LEARNED** Test in the final size. A small gap that is invisible on a screenshot shows up on a big screen.
2	**My sliders did not update**	**PROBLEM** First they were invisible, then they moved but the numbers on the page stayed still.	**WHAT I DID** The slicer used "Before" mode, which sends a range. SELECTEDVALUE returns blank for a range, so I switched to MAX in the Value measures.	**WHAT I LEARNED** Know what a slicer really sends to the model.
3	**Month over month did not work**	**PROBLEM** The growth chart looked wrong because each month could only see itself.	**WHAT I DID** My calendar was not marked as a date table, so I used REMOVEFILTERS and EDATE to look at the month before.	**WHAT I LEARNED** Sometimes the right fix is in the measure, and sometimes it is in the model. Now I know both.
4	**Chart type and label colours**	**PROBLEM** A waterfall chart would not let me colour the labels the way I wanted.	**WHAT I DID** I switched to a clustered column chart and used two colour measures, one for the bar and one for the label.	**WHAT I LEARNED** Pick the chart type that gives me control, not the one that looks cleverest.
5	**The JSON theme was rejected**	**PROBLEM** Power BI refused my first theme file because of a property it does not accept.	**WHAT I DID** I removed the shadow property and kept the colours, fonts and styles. That became my "no shadow" theme.	**WHAT I LEARNED** Start with a small theme and add one setting at a time.
6	**Labels that would not scale**	**PROBLEM** K and M labels did not shrink the numbers when I used trailing commas in FORMAT.	**WHAT I DID** I divided the number myself and added the K or M as text.	**WHAT I LEARNED** FORMAT is for display, not for scaling.
7	**Hundreds of objects on a page**	**PROBLEM** It was easy to move the wrong thing by accident, and impossible to find a chart in the selection pane.	**WHAT I DID** I renamed and grouped every object by purpose, and checked that nothing changed place.	**WHAT I LEARNED** Always rename your selections. Good names are part of the design.

More technical challenges and what I learned

These are the problems I met while building, and what each one taught me.

8	**Data preparation and saving my work**	**PROBLEM** In Power Query I deleted and kept columns and rows, and I was not always sure the next step was safe.	**WHAT I DID** I saved a copy of my file before every big change, so I always had two files. When something broke, I copied the last good file and started again from there.	**WHAT I LEARNED** Save a copy before you change the source. It is the cheapest way to be safe.
9	**Formulas that were not allowed**	**PROBLEM** Some DAX formulas I typed were not allowed, and I had to redo them.	**WHAT I DID** I started writing them in small steps. VAR holds each part and RETURN gives the answer.	**WHAT I LEARNED** Small steps are easier to fix than one long formula.
10	**The background kept popping up**	**PROBLEM** When I adjusted a text box, the background shape kept popping up and covering what I was working on.	**WHAT I DID** I opened the Selection pane from the View tab and picked the one object I wanted, so I could focus on it.	**WHAT I LEARNED** Use the Selection pane to work on one object at a time.
11	**A line under my text box**	**PROBLEM** I put a DAX measure inside a text box, but an underline kept showing up.	**WHAT I DID** I used a Card visual instead and made its background transparent.	**WHAT I LEARNED** If one visual fights me, I try another visual that does the same job.
12	**Filtering by category name did not work**	**PROBLEM** For the category performance and action table, filtering by the category name was not working.	**WHAT I DID** I filtered with the Product Category Key instead. Key number 1 keeps only the bikes.	**WHAT I LEARNED** Filter on the key, not on the words.

MY FAVOURITE CHALLENGE: TOGGLE BUTTONS

Toggle buttons were hard and also my favourite. I used bookmarks and hidden slicers, connected them so they work by themselves, and renamed every selection so I could find it later. One toggle lets a page show two charts and two analyses, so people can compare them easily. It took effort, because I really wanted my dashboard to feel like an app, with good UI and UX.

WHAT I ENJOYED: STACKED BARS AND COLUMNS

I love stacked bar and stacked column charts for showing progress and for comparing two sets of data. I used DAX measures for the track gap, so each bar shows how far it has come and how far it can go.

What this project taught me

- **Plan the question before the chart.** The twelve questions saved me from adding charts that looked nice but meant nothing.
- **Write measures in small steps.** VAR boxes made my long formulas easy to read and fix.
- **Save a copy before big changes.** Two files saved me when something broke.
- **Keep the model small.** Fewer columns and more measures made everything faster.
- **Design is a system.** A theme file, one grid and one set of names made six pages feel like one product.
- **Always check against the real numbers.** I compared every figure on the page to the model before I trusted it.
- **Name and group as I go.** Cleaning up 255 measures and hundreds of visuals at the end took a lot of time.
- **Say what a number means.** A sentence under a chart did more than another chart.
- **Say what I am not sure about.** My forecast has a range, and my actions are estimates.

12
business questions asked first

6
pages, one question each

6
actions with a value

1
theme file for the whole look

My checklist before I call a page finished

- ✓ Every number matches the model, and the text under each chart matches the number above it.
- ✓ The date picker changes every card, chart and sentence on the page.
- ✓ Every button and slider works, and the menu looks the same on all six pages.
- ✓ Nothing is cut off. Long words and numbers fit inside their cards.
- ✓ Colours follow the theme: blue for focus, green and red only for good and bad.
- ✓ Every object has a name and sits in a group, so nothing moves by accident.

WHAT I WOULD DO NEXT TIME

I would mark my calendar as a date table from the start, move measures into their tables right when I write them, and save a theme file after every change so I can go back.

17 Impact and value

What this project gives to a business, and to me

A leader of Miles & Memories can open one file and, in a few minutes, know how the company is doing, who matters, where to act and what it is worth.

Value to the business

$2.01M
revenue opportunity found in three gaps

$200.07K
extra gross profit from a better mix

$502.75K
revenue protected in the top decile

1
trusted page instead of many reports

- **Faster decisions.** Every page ends in a recommendation, so a meeting can start from "what do we do?" and not from "what are the numbers?".
- **One version of the truth.** All numbers come from one model and one set of measures.
- **Margin on the first page.** Leaders see revenue and profit together, so growth is never judged alone.
- **Easy to keep fresh.** New sales files go into the folder, and a refresh updates every page.

Skills I showed in this project

Skill	What I did
Data preparation	Power Query: data types, text clean up, a combined folder of files, helper columns
Data modelling	Star schema with a snowflake branch, six relationships, disconnected helper tables
DAX	255 measures: VAR and RETURN, CALCULATE, iterators, time intelligence, ranking, a straight line forecast
Analysis	Descriptive, diagnostic, predictive and prescriptive, in that order
Design	JSON theme, a SaaS style layout, navigation, toggles, sliders, conditional colours and reference lines
Storytelling	A question on every page, a "What this means" panel and a recommendation with a value

What I would add next

- A live data connection so the dashboard refreshes by itself.
- A forecast that also learns the seasons, such as the December spike.
- Row level security, so each country manager sees only their market.
- A phone layout for leaders who check numbers on the go.

Data is only useful when it ends in a decision.

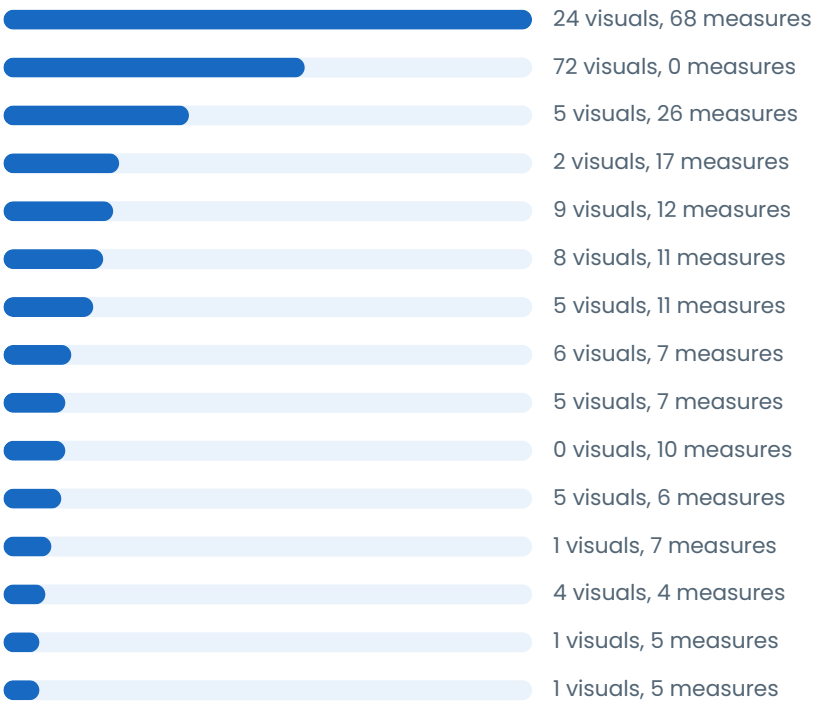
EVERY MILE BECOMES A MEMORY.

A Appendix A: all 255 DAX measures

From most important to least important, grouped by the table they live in

I ranked each measure by how much the dashboard depends on it: **one point for every visual that uses it** and **one and a half points for every other measure that is built on top of it**. A measure that many things depend on is more important than one that sits on a single card.

The 15 measures the dashboard depends on most



Where they live

Table	Measures	What lives there
_Measure	124	Base KPIs, colour rules, text and captions, and the readers for buttons.
_Calculate	84	Calculations with logic: gaps, baskets, segments, rankings and the Strategy numbers.
_TIME INT	37	Anything that compares periods: trailing, growth, half year, trend and forecast.
Helper tables	10	Helper measures that sit in the parameter or selector table they belong to.

HOW TO READ THE LISTS

Pages shows where the measure ends up (Ov Overview, Pr Products, Cu Customers, Te Territory, Tr Trends, St Strategy), including through other measures. **Visuals** is the number of visuals on the six pages that use the measure directly. **Built on it** is the number of other measures that call it. A zero in both columns means the measure is a leftover. There are 17 of them, and I can delete them to leave 238 active measures.

TABLE

_Measure

Base KPIs, colour rules, text and captions, and the readers for buttons.

124

m

#	Measure	What it does	Pages	Visuals	Built on
1	**Active Page**	Tells each sidebar button which page is open, so the open page lights up.	All	72	0
2	**Total Orders**	Counts each order number once.	Ov Cu Te Tr St	5	26
3	**Average Order Value (AOV)**	Revenue divided by the number of orders.	Ov Pr Te St	2	17
4	**Profit Margin %**	Total Profit divided by Total Revenue.	Ov Pr Te St	9	12
5	**Total Profit**	Total Revenue minus Total Cost. This is my gross profit.	Ov Pr Te St	8	11
6	**Total Customers**	Counts each customer once.	Ov Cu Te	6	7
7	**Total Unit Sold**	Adds up the quantity sold.	Pr	5	7
8	**Selected Metric**	Reads the Revenue, Profit or Orders button on the Overview and Trends charts.	Ov Tr	0	10
9	**Revenue per Customer**	Revenue divided by customers.	Ov Cu Te	5	6
10	**Active View**	Reads which toggle button is pressed (All, Bikes or Accessories).	Pr Te	6	0
11	**Greeting Date**	Shows today's date and time under the greeting.	All	6	0
12	**Greeting Text**	Says good morning, afternoon or evening, based on the time.	All	6	0
13	**Strat Weak Market**	Finds the market with the smallest basket.	St	0	4
14	**Total Products**	Counts the different products that were sold.	Pr	4	1
15	**H1 Growth Display**	Card text for half year growth, with an arrow and a sign (like ▲ 7.91%).	Pr Tr	4	0
16	**AOV Display**	Card text for average order value, with an arrow and a sign (like ▲ 7.91%).	Te	3	0
17	**Action Tag**	Shows the action word (Promote, Reprice and so on) for a category or subcategory.	Pr	3	0
18	**H1 Growth Color**	Colour rule for half year growth: green for good, red for bad.	Pr	3	0
19	**Territory Call**	Gives each country a call: Invest, Maintain or Build basket.	Te	3	0
20	**Territory Growth Display**	Card text for territory growth, with an arrow and a sign (like ▲ 7.91%).	Te	3	0
21	**Trend AOV Delta**	Average order value of the last three months minus the three before.	Ov	0	2
22	**Trend Margin Delta Pts**	Margin of the last three months minus the three before, in points.	Ov	0	2
23	**Repeat Customer Rate %**	Share of customers who ordered more than once.	Ov Cu	1	1
24	**Total Revenue (Components)**	Revenue from components.	Ov	1	1
25	**Active Segment View**	Reads which toggle button is pressed (Revenue or Count).	Cu	2	0
26	**Segment Play**	Gives each customer segment its play: Retain, Win back, Nurture, Reward or Reactivate.	Cu	2	0
27	**Top Country Pill Right**	Text in the pill for top country.	Ov Te	2	0
28	**Category Action**	Gives each category its action: Reprice, Promote, Expand or Dormant.	Pr	0	1
29	**Strat Best Market**	Finds the market with the biggest basket.	St	0	1
30	**Subcategory Action**	Gives each subcategory an action based on its revenue and margin.	Pr	0	1
31	**Total Cost**	Quantity sold times the product cost, added up.	Ov Pr Te St	0	1
32	**Trend Streak Callout**	Sentence about trend streak at the bottom of its card, written from live numbers.	Tr	0	1

#	Measure	What it does	Pages	Visuals	Built on
33	**AOV Caption**	Small caption under the average order value number. It changes with the filters.	Ov	1	0
34	**Basket Effect Callout**	Sentence about basket effect at the bottom of its card, written from live numbers.	Ov	1	0
35	**Basket Gap Callout**	Sentence about basket gap at the bottom of its card, written from live numbers.	Te	1	0
36	**Basket Gap Display**	Card text for basket gap, with an arrow and a sign (like ▲ 7.91%).	Te	1	0
37	**Basket Uplift Color**	Colour rule for basket uplift: green for good, red for bad.	Ov	1	0
38	**Basket Uplift Label**	Label for basket uplift, so the text follows the data.	Ov	1	0
39	**Biggest Basket Caption**	Small caption under the biggest basket number. It changes with the filters.	Te	1	0
40	**Bike Only Margin Caption**	Small caption under the bike only margin number. It changes with the filters.	Ov	1	0
41	**Bike Plus Accessory Margin Caption**	Small caption under the bike plus accessory margin number. It changes with the filters.	Ov	1	0
42	**Bridge Label Color**	Colour rule for the bridge label: the highlight colour goes to the top item.	Tr	1	0
43	**Category Driver**	Labels a category as a revenue driver or a margin driver.	Pr	1	0
44	**Fastest Growth Caption**	Small caption under the fastest growth number. It changes with the filters.	Te	1	0
45	**Fastest Growth Color**	Colour rule for fastest growth: green for good, red for bad.	Te	1	0
46	**Fastest Growth Display**	Card text for fastest growth, with an arrow and a sign (like ▲ 7.91%).	Te	1	0
47	**Gross Profit Caption**	Small caption under the gross profit number. It changes with the filters.	Ov	1	0
48	**Group Avg Spend**	Average spend for the customer group picked (home owners, renters, degree holders).	Cu	1	0
49	**Group Customers**	Number of customers in the customer group picked.	Cu	1	0
50	**Growth Second Color**	Colour rule for growth second: green for good, red for bad.	Te	1	0
51	**Growth Second Display**	Card text for growth second, with an arrow and a sign (like ▲ 7.91%).	Te	1	0
52	**Growth Second Name**	Name of the item for growth second, read from the data.	Te	1	0
53	**Growth Slowest Color**	Colour rule for growth slowest: green for good, red for bad.	Te	1	0
54	**Growth Slowest Display**	Card text for growth slowest, with an arrow and a sign (like ▲ 7.91%).	Te	1	0
55	**Growth Slowest Name**	Name of the item for growth slowest, read from the data.	Te	1	0
56	**Growth Source Callout**	Sentence about growth source at the bottom of its card, written from live numbers.	Tr	1	0
57	**Income Band Callout**	Sentence about income band at the bottom of its card, written from live numbers.	Cu	1	0
58	**Latest Month Growth Display**	Card text for latest month growth, with an arrow and a sign (like ▲ 7.91%).	Tr	1	0
59	**Latest Month Note**	Short note about latest month, written from live numbers.	Tr	1	0
60	**Margin Callout**	Sentence about margin at the bottom of its card, written from live numbers.	Pr	1	0
61	**Margin Spotlight**	Spotlight sentence about margin, written from live numbers.	Pr	1	0
62	**MoM Fall Callout**	Sentence about month over month fall at the bottom of its card, written from live numbers.	Tr	1	0

#	Measure	What it does	Pages	Visuals	Built on
63	**MoM Fill Color**	Colour rule for the month over month fill: the highlight colour goes to the top item.	Tr	1	0
64	**MoM Growth % Display**	Card text for month over month growth %, with an arrow and a sign (like ▲ 7.91%).	Ov	1	0
65	**MoM Growth Caption**	Small caption under the month over month growth number. It changes with the filters.	Ov	1	0
66	**MoM Label Color**	Colour rule for the month over month label: the highlight colour goes to the top item.	Tr	1	0
67	**Order Growth Display**	Card text for order growth, with an arrow and a sign (like ▲ 7.91%).	Tr	1	0
68	**Order Growth Note**	Short note about order growth, written from live numbers.	Tr	1	0
69	**Orders per Customer**	Orders divided by customers, to show repeat buying.	Te	1	0
70	**Orders per Customer Caption**	Small caption under the orders per customer number. It changes with the filters.	Cu	1	0
71	**RPC Callout**	Sentence about rpc at the bottom of its card, written from live numbers.	Te	1	0
72	**Repeat Rate Caption**	Small caption under the repeat rate number. It changes with the filters.	Ov	1	0
73	**Returning Orders Caption**	Small caption under the returning orders number. It changes with the filters.	Cu	1	0
74	**Revenue Share Caption (Accessories)**	A small caption under a number. It updates when the filters change.	Pr	1	0
75	**Revenue by Category Callout**	Sentence about revenue by category at the bottom of its card, written from live numbers.	Ov	1	0
76	**Scatter Dot Color**	Colour rule for the scatter dot: the highlight colour goes to the top item.	Pr	1	0
77	**Scatter Footnote**	Footnote for scatter that writes itself from the data.	Pr	1	0
78	**Seasonality Bar Color**	Colour rule for the seasonality bars: the highlight colour goes to the top item.	Tr	1	0
79	**Seasonality Label Color**	Colour rule for the seasonality label: the highlight colour goes to the top item.	Tr	1	0
80	**Seasonality Peak Display**	Card text for seasonality peak, with an arrow and a sign (like ▲ 7.91%).	Tr	1	0
81	**Seasonality Peak Text**	Sentence about seasonality peak, written from live numbers.	Tr	1	0
82	**Seasonality Trough Display**	Card text for seasonality trough, with an arrow and a sign (like ▲ 7.91%).	Tr	1	0
83	**Seasonality Trough Text**	Sentence about seasonality trough, written from live numbers.	Tr	1	0
84	**Segment Table Note**	Short note about segment table, written from live numbers.	Cu	1	0
85	**Smallest Basket Caption**	Small caption under the smallest basket number. It changes with the filters.	Te	1	0
86	**Strat Action 01**	The 'what to do' sentence on a strategy card, built from my data.	St	1	0
87	**Strat Action 02**	The 'what to do' sentence on a strategy card, built from my data.	St	1	0
88	**Strat Action 04**	The 'what to do' sentence on a strategy card, built from my data.	St	1	0
89	**Strat Evidence 01**	The evidence sentence on a strategy card, built from my data.	St	1	0
90	**Strat Evidence 02**	The evidence sentence on a strategy card, built from my data.	St	1	0
91	**Strat Evidence 03**	The evidence sentence on a strategy card, built from my data.	St	1	0
92	**Strat Evidence 04**	The evidence sentence on a strategy card, built from my data.	St	1	0

#	Measure	What it does	Pages	Visuals	Built on
93	**Strat Evidence 05**	The evidence sentence on a strategy card, built from my data.	St	1	0
94	**Strat Evidence 06**	The evidence sentence on a strategy card, built from my data.	St	1	0
95	**Strat Hero Note**	The paragraph in the blue box that sums up the total opportunity.	St	1	0
96	**Strat Pill 01**	The blue pill text on a strategy card. It changes when I move the slider.	St	1	0
97	**Strat Pill 02**	The blue pill text on a strategy card. It changes when I move the slider.	St	1	0
98	**Strat Pill 03**	The blue pill text on a strategy card. It changes when I move the slider.	St	1	0
99	**Strat Pill 04**	The blue pill text on a strategy card. It changes when I move the slider.	St	1	0
100	**Strat Pill 05**	The blue pill text on a strategy card. It changes when I move the slider.	St	1	0
101	**Strat Slider Text 01**	The small line under a slider, such as 100% of the gap applied.	St	1	0
102	**Strat Slider Text 02**	The small line under a slider, such as 100% of the gap applied.	St	1	0
103	**Strat Slider Text 03**	The small line under a slider, such as 100% of the gap applied.	St	1	0
104	**Strat Slider Text 04**	The small line under a slider, such as 100% of the gap applied.	St	1	0
105	**Strat Upside Label Color**	Colour rule for the strat upside label: the highlight colour goes to the top item.	St	1	0
106	**Table Note**	Short note about table, written from live numbers.	Pr	1	0
107	**Territory Note**	Short note about territory, written from live numbers.	Te	1	0
108	**Top Products Callout Text**	Sentence about top products callout, written from live numbers.	Ov	1	0
109	**Top Products Subtitle**	Subtitle for top products that writes itself from the data.	Ov	1	0
110	**Top Rider Caption**	Small caption under the top rider number. It changes with the filters.	Cu	1	0
111	**Top Riders Callout**	Sentence about top riders at the bottom of its card, written from live numbers.	Cu	1	0
112	**Trend Best Month Caption**	Small caption under the trend best month number. It changes with the filters.	Ov	1	0
113	**Trend Chart Subtitle**	Subtitle for trend chart that writes itself from the data.	Ov	1	0
114	**Trend Metric Callout**	Sentence under the Trends chart about the winning streak and the projection, for the chosen button.	Tr	1	0
115	**Trend MoM Growth Color**	Colour rule for trend month over month growth: green for good, red for bad.	Ov	1	0
116	**Trend MoM Growth Display**	Card text for trend month over month growth, with an arrow and a sign (like ▲ 7.91%).	Ov	1	0
117	**Trend Sales Subtitle**	Subtitle for trend sales that writes itself from the data.	Tr	1	0
118	**Trend Third Stat Color**	Colour rule for trend third stat: green for good, red for bad.	Ov	1	0
119	**Trend Third Stat Display**	Card text for trend third stat, with an arrow and a sign (like ▲ 7.91%).	Ov	1	0
120	**Trend Third Stat Label**	Label for trend third stat, so the text follows the data.	Ov	1	0
121	**Average Unit Revenue**	Revenue divided by units sold.	-	0	0
122	**Continent Bar Color**	Colour rule for the continent bars: the highlight colour goes to the top item.	-	0	0
123	**Trend Best Month Display**	Card text for trend best month, with an arrow and a sign (like ▲ 7.91%).	-	0	0

TABLE

_Calculate

Calculations with logic: gaps, baskets, segments, rankings and the Strategy numbers.

84 measures

#	Measure	What it does	Pages	Visuals	Built on
1	**Total Revenue**	Quantity sold times the product price, added up. The base of almost everything.	All	24	68
2	**Income Band Avg Spend**	Average spend per customer in the income band.	Cu	4	4
3	**Company AOV**	The company average order value, ignoring the market filter.	Te St	1	5
4	**Top Product by Revenue**	Name of the product with the highest revenue.	Pr	1	5
5	**Decile Revenue Share %**	Share of all revenue from each customer decile.	Cu	2	4
6	**Grand Total Revenue**	All revenue with every filter removed. Used as the 100% line.	Ov Pr	0	5
7	**Income Band Customers**	Number of customers in the income band.	Cu	3	3
8	**Top Country Name**	Name of the country with the most revenue.	Ov Te	2	3
9	**Strat Attach Gap $**	Revenue gained if bike-only orders also bought an average accessory order.	St	0	4
10	**Income Band Revenue**	Revenue from the income band.	Cu	2	2
11	**Bike Only Margin %**	Margin of bike-only orders.	Ov St	0	3
12	**Bike Plus Accessory Margin %**	Margin of bike plus accessory orders.	Ov St	0	3
13	**Strat Basket Gap $**	Revenue gained if the weakest market reached the company average basket.	St	0	3
14	**Strat Reactivation $**	Revenue gained if 5% of lapsing riders come back and spend again.	St	0	3
15	**Basket Uplift $**	Difference between the two average order values.	Ov	1	2
16	**Decile Track Gap**	Same trick for deciles: decile one gets its own colour.	Cu	1	2
17	**Income Band Track Gap**	Same trick for income bands: the top band gets its own colour.	Cu	1	2
18	**Lowest Margin Product**	Name of the product with the lowest margin.	Pr	1	2
19	**Occupation Track Gap**	Same trick for occupations: the top spender gets its own colour.	Cu	1	2
20	**Strat At Risk $**	Revenue at risk if 5% of the top decile stops buying.	St	1	2
21	**Total Revenue (Accessories)**	Revenue from accessories.	Ov Pr	1	2
22	**Total Revenue (Apparel)**	Revenue from clothing.	Ov	1	2
23	**New Customers**	Customers whose first order falls in the period.	Ov Cu	2	1
24	**Segment Median Recency**	Middle number of days since last order in a segment.	Cu	2	1
25	**Pct Revenue Cont (ALLSELECTED Pro)**	Share of the visible products' revenue, for the table bars.	Pr	3	0
26	**Strat Mix Shift Profit $**	Profit gained if 25% of road bike revenue moves to mountain bikes.	St	0	2
27	**Total Revenue (Bikes)**	Revenue from bikes.	Ov	0	2
28	**Biggest Basket AOV**	Highest average order value among the markets.	Te	1	1
29	**Bike Attach Rate %**	Share of bike orders that also include an accessory.	Ov St	1	1
30	**Bike Only AOV**	Average order value of bike-only orders.	Ov	1	1
31	**Bike Plus Accessory AOV**	Average order value of bike plus accessory orders.	Ov	1	1
32	**Bridge Value**	Feeds the What built the growth waterfall: H1 2022, each category, then total.	Tr	1	1
33	**Margin % (Accessories)**	Margin of accessories.	Ov Pr	1	1
34	**Returning Customer Orders**	Orders minus new customers, so the repeat orders.	Cu	1	1

#	Measure	What it does	Pages	Visuals	Built on
35	**Revenue Split (Top Quartile)**	The revenue level that marks the top quarter of products (scatter line).	Pr	1	1
36	**Scatter Y Split**	The middle value for the scatter Y line (margin or volume).	Pr	1	1
37	**Smallest Basket AOV**	Lowest average order value among the markets.	Te	1	1
38	**Strat Revenue Upside**	The three revenue actions added up, scaled by the three sliders.	St	1	1
39	**Strat Upside Value**	Feeds the upside waterfall: today's revenue, then each action as a step.	St	1	1
40	**Top Decile Share %**	Share of revenue from the top 10% of customers.	Cu St	1	1
41	**% of Total Revenue**	Revenue as a share of all revenue.	Pr	0	1
42	**Adjusted Revenue**	Revenue after a price change from the what-if slider.	-	0	1
43	**Margin % (Bikes)**	Margin of bikes.	Ov	0	1
44	**Revenue Share % (Accessories)**	Category revenue as a share of all revenue.	Pr	0	1
45	**Revenue Share % (Bikes)**	Category revenue as a share of all revenue.	Ov	0	1
46	**Top 10 Model Count**	How many different models sit inside the top ten products.	Ov	0	1
47	**Top 10 Revenue Share %**	Share of revenue from the ten best products.	Ov	0	1
48	**Top Country Share %**	Share of revenue from the top country.	Ov Te	0	1
49	**Accessories Revenue**	Revenue from accessories, used on the donut legend.	Ov	1	0
50	**Basket Flag**	Gives 1 if the market basket is above the company average.	Te	1	0
51	**Bike Revenue**	Revenue from bikes, used on the donut legend.	Ov	1	0
52	**Decile Other Share**	The normal bar value. It is filled for every item except the top one.	Cu	1	0
53	**Decile Top Share**	The highlighted bar value. It is filled only for the top item.	Cu	1	0
54	**Income Band Other Spend**	The normal bar value. It is filled for every item except the top one.	Cu	1	0
55	**Income Band Top Spend**	The highlighted bar value. It is filled only for the top item.	Cu	1	0
56	**Is Top Country**	Gives 1 for the top country and 0 for the rest, to colour its bar.	Ov	1	0
57	**Is Top Product**	Gives 1 for the top product and 0 for the rest.	Pr	1	0
58	**Is Top RPC Market**	Gives 1 for the market with the highest revenue per customer.	Te	1	0
59	**Lowest Margin Product Margin %**	Margin of the lowest margin product.	Pr	1	0
60	**Lowest Margin Product Units**	Units sold of the lowest margin product.	Pr	1	0
61	**Occupation Other Spend**	The normal bar value. It is filled for every item except the top one.	Cu	1	0
62	**Occupation Top Spend**	The highlighted bar value. It is filled only for the top item.	Cu	1	0
63	**Other Continent Revenue**	Revenue of the other continents, for the colour rule.	Te	1	0
64	**Scatter Y Value**	Gives the scatter chart its Y value: margin or volume.	Pr	1	0
65	**Segment Rider Share %**	Share of all riders that sit in the segment.	Cu	1	0
66	**Strat Profit Upside**	The profit action, scaled by its slider.	St	1	0
67	**Subcategories Selling**	Counts subcategories with sales.	Pr	1	0
68	**Top Continent Revenue**	Revenue of the best continent, for the colour rule.	Te	1	0
69	**Top Country AOV**	Average order value of the top country.	Te	1	0
70	**Top Country Revenue**	Revenue of the top country.	Te	1	0
71	**Top Product Margin %**	Margin of the top product.	Pr	1	0

#	Measure	What it does	Pages	Visuals	Built on
72	**Top Product Revenue**	Revenue of that top product.	Pr	1	0
73	**Top Product Share %**	Share of revenue from the top product.	Pr	1	0
74	**Top Product Units**	Units sold of the top product.	Pr	1	0
75	**Top Two Deciles Share %**	Share of revenue from the top 20% of customers.	Cu	1	0
76	**Total Units Sold (Accessories)**	Units sold of accessories.	Pr	1	0
77	**Track Gap**	Gap between the best continent and this one. It makes the best bar a different colour.	Te	1	0
78	**Margin % (Apparel)**	Margin of clothing.	-	0	0
79	**Markets Below Avg Basket**	Counts markets with a basket below the company average.	-	0	0
80	**Projected Attach Revenue**	Revenue if more bike orders attached an accessory.	-	0	0
81	**Revenue Impact of Price Change**	Adjusted revenue minus real revenue.	-	0	0
82	**Revenue Share % (Apparel)**	Category revenue as a share of all revenue.	-	0	0
83	**Revenue Share % (Components)**	Category revenue as a share of all revenue.	-	0	0
84	**Total Products (Bikes)**	Number of bike products sold.	-	0	0

TABLE

_TIME INT

Anything that compares periods: trailing, growth, half year, trend and forecast.

37
measures

#	Measure	What it does	Pages	Visuals	Built on
1	**Trailing 3-Month Growth %**	Growth of the last three months against the three before.	Ov Te	5	11
2	**Trailing 3M Revenue**	Revenue of the last three months.	Ov Te Tr	0	4
3	**H1 Growth %**	Growth of the latest H1 against the prior H1.	Pr Tr	1	3
4	**H1 Revenue Prior**	Revenue from January to June of the year before.	Tr	1	3
5	**Latest Month Revenue**	Revenue of the last month in the file.	Tr	1	3
6	**Latest Month Orders**	Orders of the last month in the file.	Tr	2	2
7	**Prev. Month Revenue**	Revenue of the month before.	Ov	0	3
8	**Prior 3M Revenue**	Revenue of the three months before that.	Ov Te	0	3
9	**H1 Revenue Latest**	Revenue from January to June of the latest year.	Tr	1	2
10	**MoM Chart Growth %**	Month over month growth made for the chart. It keeps working with the date filter.	Tr	1	2
11	**Trend Best Month Value**	Revenue or orders of the best month.	Ov	1	2
12	**Trend Metric Projection**	The dashed straight line projection for July to September, for the chosen button.	Tr	2	1
13	**Trend MoM Growth 3M Avg**	Average month over month growth of the last three months.	Ov	0	2
14	**Latest Month Growth %**	Growth of the last month against the month before.	Tr	1	1
15	**MoM Growth %**	Revenue growth against the previous month.	Ov	1	1
16	**Trend Metric 3M Avg**	The light dashed three month average on the Trends chart, for the chosen button.	Tr	2	0
17	**Trend Metric Actual**	The solid line on the Trends chart. It follows the Revenue or Orders button.	Tr	2	0
18	**Actual Sales**	Monthly revenue for real months (solid line).	Tr	0	1
19	**Forecast Sales**	Projected revenue for the next three months (dashed line).	Tr	0	1
20	**H1 Revenue Change**	Latest H1 minus prior H1.	Tr	0	1
21	**Prior 3M Orders**	Orders of the three months before that.	Ov	0	1
22	**Prior 3M Profit**	Profit of the three months before that.	Ov	0	1

#	Measure	What it does	Pages	Visuals	Built on
23	**Trailing 3M Orders**	Orders of the last three months.	Ov	0	1
24	**Trailing 3M Profit**	Profit of the last three months.	Ov	0	1
25	**Trend 3M Avg Revenue**	Three month average line on the trend chart.	Tr	0	1
26	**Trend Best Month Date**	The date of the best month.	Ov	0	1
27	**First Month AOV**	Average order value of the first month in the file.	Tr	1	0
28	**Latest Month AOV**	Average order value of the last month.	Tr	1	0
29	**Seasonality Avg Revenue**	Average revenue of the months shown. It is the dashed red line.	Tr	1	0
30	**Trailing 3M Avg Revenue**	Trailing 3 month revenue divided by 3.	Tr	1	0
31	**MoM Worst Fall %**	The biggest real monthly fall in the actual data.	-	0	0
32	**Prev. Month Profit**	Profit of the month before.	-	0	0
33	**Prev. Month Units Sold**	Units sold in the month before.	-	0	0
34	**SPLY Revenue**	Revenue in the same period last year.	-	0	0
35	**Year-to-Date**	Revenue from January 1 to the date.	-	0	0
36	**vs Prev. Month Revenue ($)**	Change in dollars against the previous month.	-	0	0
37	**vs Prev. Month Revenue (%)**	Change in percent against the previous month.	-	0	0

TABLE

Helper measures in their own tables

10 measures

Helper measures that sit in the parameter or selector table they belong to.

#	Measure	What it does	Pages	Visuals	Built on
1	**Trend Bar Value** (Metric Selector)	Gives the bar or line its value: revenue, profit or orders, based on the button.	Ov Tr	1	7
2	**Attach Gap Slider Value** (Attach Gap Slider)	Reads the position of the attach gap slider (0 to 100).	St	0	4
3	**Canada Basket Slider Value** (Canada Basket Slider)	Reads the position of the Canada slider (0 to 100).	St	0	4
4	**Reactivation Slider Value** (Reactivation Slider)	Reads the position of the reactivation slider (0 to 100).	St	0	4
5	**Mix Shift Slider Value** (Mix Shift Slider)	Reads the position of the mix shift slider (0 to 100).	St	0	3
6	**Selected Scatter Mode** (Metric Selector)	Reads the Margin or Volume button on the product scatter chart.	Pr	0	3
7	**Bike Attach Rate Uplift % Value** (Bike Attach Rate Uplift %)	Reads the attach rate lift picked on the what-if slider.	-	0	1
8	**Price Adjustment % Value** (Price Adjustment %)	Reads the price change picked on the what-if slider.	-	0	1
9	**Trend Orders 3M Avg** (Metric Selector)	Three month average of orders, for the Orders button.	Ov	0	1
10	**Trend Line Value** (Metric Selector)	Gives the line its value: the 3 month average for the metric picked.	Ov	1	0

B Appendix B: Words I used

A short list in plain English

Word	Meaning
AOV	Average order value. Revenue divided by the number of orders.
Attach rate	The share of bike orders that also include an accessory.
Bookmark	A saved view of a page. My buttons use bookmarks to switch views.
Calculated column	A column made with a formula. It is stored for every row.
DAX	The formula language of Power BI.
Decile	One tenth of the customers, ranked by how much they spend.
Dimension table	A table that describes things, such as products or customers.
Fact table	A table of events, such as one row for each sale.
Field parameter	A Power BI table that lets a viewer change the field used in a chart.
Gross profit	Revenue minus the cost of the products.
JSON theme	A text file with the colours and styles Power BI applies to every chart.

Word	Meaning
KPI	A key number that tells how the business is doing.
Margin	Profit divided by revenue.
Measure	A formula worked out when a chart asks for it. It does not take space.
MoM	Month over month. This month against the month before.
Power Query	The tool that loads and cleans data before it goes into the model.
Relationship	A link between two tables, made with a key.
RFM	Recency, frequency and monetary value, used to group customers.
Snowflake	A dimension that connects to another dimension, like Products to Subcategories to Categories.
Star schema	One fact table in the middle with dimension tables around it.
Trailing 3 months	The latest three months, compared with the three months before.
What-if parameter	A slider that lets a viewer test a different assumption.

Files that come with this project

- **Miles and Memories Executive Dashboard.pbix**: the final Power BI report.
- **miles-memories-theme.json**: my JSON theme.
- **Measure move checklist**: the list I used to place measures in their tables.

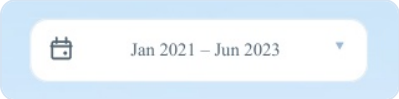
Cristine Angelica Marie Serrano Tolentino
Business Intelligence Analyst | Data Analyst

Every Mile Becomes a Memory.

C Appendix C: What changed in this version

Updates made after the first release of the dashboard

1. The date picker is now a pill with a month list



On all six pages the date picker is a white pill that shows the range in view, such as **Jan 2021 – Jun 2023**. Click the pill and a list of months drops down under it, one row per month from Jan 2021 to Jun 2023. Tick the months you want and every card, chart and sentence on the page recalculates. The text in the pill changes with the choice, for example Jul 2022 – Jun 2023.

It is built like my Revenue and Orders toggle: the pill text is a card, two invisible buttons open and close the list, and bookmarks show or hide it without touching the date choice.

What went wrong first. My first list used the Date column, so each month appeared about 30 times (once per day). I changed the list to `Dim_Calendar[Start of Month]`, so each month shows once.

New measure, `Date Range Label` in _Measure (it is the 255th):

```
VAR StartDate = MIN ( Dim_Calendar[Date] )
VAR EndDate = MAX ( Dim_Calendar[Date] )
VAR StartText = FORMAT ( StartDate, "mmm yyyy" )
VAR EndText = FORMAT ( EndDate, "mmm yyyy" )
RETURN
    IF ( StartText = EndText, StartText, StartText & " – " & EndText )
```

2. The basket effect now follows the dates

The problem. The basket measures read my `Order Basket` table. That table is built once, when the data loads, from every date, and it has no link to the calendar. So the date list could not filter it, and the basket effect never moved.

The fix. Eight measures now build the same list of orders on the spot from `Fact_Sales`, using only the dates that are picked. They keep their names, so no visual had to change: Bike Only AOV, Bike Plus Accessory AOV, Bike Only Margin %, Bike Plus Accessory Margin %, Bike Attach Rate %, Strat Attach Gap $, Strat Evidence 02 and Basket Effect Callout.

```
VAR Orders = ADDCOLUMNS ( SUMMARIZE ( Fact_Sales, Fact_Sales[OrderNumber] ),
    "@HasBike", CALCULATE ( COUNTROWS ( Fact_Sales ), Dim_Product_Categories[CategoryName] = "Bikes" ) > 0,
    "@HasAccessory", CALCULATE ( COUNTROWS ( Fact_Sales ), Dim_Product_Categories[CategoryName] = "Accessories" ) > 0 )
VAR BikeOrders = COUNTROWS ( FILTER ( Orders, [@HasBike] ) )
VAR AttachedOrders = COUNTROWS ( FILTER ( Orders, [@HasBike] && [@HasAccessory] ) )
RETURN DIVIDE ( AttachedOrders, BikeOrders )     // Bike Attach Rate %
```

How I checked it. With every month picked, the result is the same as before. With a range picked, it moves (counts from the sales table):

Dates picked	Bike only orders	Bike + accessory orders	Attach rate
Jan 2021 – Jun 2023 (all)	6,820	7,109	51.04%
Year 2021	2,630	0	0.00%
Year 2022	2,702	2,908	51.84%
Jan 2023 – Jun 2023	1,488	4,201	73.84%

WHY 2021 SHOWS ZERO

Accessories launched in July 2022, so a range inside 2021 has no bike + accessory orders. The Basket Effect Callout then says so in a sentence instead of showing an empty number.

3. What this does to the counts

- The model now has **255 measures**: 124 in _Measure, 84 in _Calculate, 37 in _TIME INT and 10 helper measures.
- 238 are used by a visual or another measure, and 17 are still leftovers.
- Appendix A was ranked before this version. `Date Range Label` is not in that list; it is used on all six pages, so it would sit near the top of the _Measure group. The eight rewritten measures keep their names, so their Appendix A entries still hold.
- The toggles (Revenue, Profit, Orders and the others) now use the same blue and light-blue buttons on every page.
- The `Order Basket` table is still in the model but no measure reads it now, so it can be deleted later.